Svaraa Lite V1 Software — PRD & Wireframes

Sep 26, 2026 · Vishwajeet J K · Svaraa Ventures Inc. · Draft v0.1 · Companion to *Svaraa Lite Hardware Specification v1.0*

Wide wireframes are set on landscape pages so nothing wraps. The live, editable version of this document is the Claude doc it was exported from.

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1. Executive summary & V1 decisions

Svaraa Lite V1 software is a web app, backed by a private per-household server, that turns the pendant's voice, button and motion events into a companion the wearer talks to and a status board the family trusts. The pendant has no internet and no lights (hardware spec §1, §7), so every internet feature, every alert and every screen lives in this software.

What V1 delivers against the seven R's

R	Wearer gets (hardware spec §3)	V1 software feature	Section
01 Recite	Mantras, bhajans, guided prayer, stories on the pendant; music on phone or speaker	Devotional library streamed to the pendant; Spotify / YouTube Music / Apple Music voice remote	12
02 Respond	Press the gold dot, speak, hear the answer	Voice assistant on the pendant + the same assistant as a chat panel in the web app	9
03 Remind	Medication, water, prayer times, appointments by voice and vibration	Reminder engine with prayer-time computation, adherence tracking, family escalation	10
04 Rescue	Hold for help; hard falls trigger "Are you okay?"	SOS and fall incident lifecycle, escalation ladder, live incident room	11
05 Recall	"What did the doctor say on Tuesday?"	Transcription, automatic session segmentation, summaries, action items, natural-language search	8
06 Repository	Private, searchable recordings and voice notes	Encrypted audio store on the household's private tenant, retention controls, export	8, 17
07 Riddle	Family, doctors, calendars, music, health connected	Connections app: calendars, health, watch, messaging channels, home hub	14

Beyond the seven R's, V1 adds a morning briefing with news (section 13), a Today dashboard (section 7), pendant management (section 15) and the onboarding that makes all of it work on day one (section 6).

Locked decisions

Decision	V1 choice	Why
Where data lives	One Svaraa-provisioned, single-tenant "Private Home Server" per household, pinned to a region the Owner picks (India default; EU, US available). Per-household data keys. Svaraa's control plane stores only account, billing and tenant routing.	Privacy promise the family can understand ("your data is on your own server"), zero setup, and Svaraa staff cannot read household data.
Who owns the account	Role-based: any adult creates a household and becomes Owner; further roles are Caregiver, Wearer, Viewer, Emergency contact. Either the elder or a family member can be Owner.	Households differ: an independent 68-year-old wants control, a 84-year-old with early dementia needs a daughter to run it.
Product surfaces	The web app (desktop and mobile browser) is 100% of the UI. A thin native "Svaraa Gateway" app on iOS/Android does only what a browser cannot: background BLE to the pendant, GPS, push, SMS/call triggers, an offline event queue. It shows the web app in a webview.	Browsers cannot keep a background BLE link (iOS Safari has no Web Bluetooth); spec §9 needs state restoration, auto-reconnect and offline queuing.
AI provider	The household picks a "Brain": Claude, OpenAI or Sarvam AI. A speech engine (STT/TTS) is paired automatically by language, with an override. Keys are Svaraa-managed by default, or bring-your-own. Switchable at any time without data loss.	Requirement; Sarvam gives the strongest Indic speech, OpenAI the widest speech stack, Claude the strongest summarisation and safety behaviour.
Recording	Record-on-request is the default with chimes (spec §7). Continuous recording is opt-in with a consent record. All audio is segmented into sessions automatically and summarised.	Lite has no recording light; the spec makes on-request the default and requires a consent prompt for continuous mode.
Emergency services	V1 alerts the family circle by push, SMS and automated call. It shows a one-tap "Call 112 / local number" to responders but never auto-dials emergency services.	Auto-dialing emergency lines is regulated and unverifiable in V1; families are the reliable first responders.
Health claims	Copy says "alerts your family after a hard fall", never "detects medical emergencies". No diagnosis, no medication advice beyond the family-entered schedule.	Spec §8.1 marketing rule; keeps V1 outside medical-device scope.

Out of scope for V1

- Bring-your-own server (a later "Advanced" path; the tenant is built so it can be exported and self-hosted).
- Native watch apps, licensed wake word, care-professional share links, an app store for third-party skills, and a dedicated home-hub mode (all placed in section 20).
- Anything the pendant cannot do: standalone internet, indoor room-level location, a visible recording indicator.

Definition of success

A household is "active" when the wearer talks to the pendant on 5 of 7 days, at least one family member opens Today on 4 of 7 days, and an SOS drill reaches a human acknowledgement in under 2 minutes.

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2. Users, roles & permissions

Five roles cover every person around one pendant, and the Wearer keeps a veto over recording and transcript access in every configuration.

Personas

Persona	Who	How they meet Svaraa	What they need
Wearer ("Amma", 74, Chennai, Tamil + English)	Wears the pendant all day; phone often charging in another room	Voice and the pendant button; opens the web app in Simple Mode on a tablet at home	Short spoken answers, prayer on time, help when she falls, family voices
Owner-Caregiver ("Priya", 42, Bengaluru, daughter)	Buys the pendant, sets everything up, pays	Web app daily on laptop and phone browser	A single Today view, reminders that stick, an SOS that reaches her in minutes
Remote Caregiver ("Ravi", 45, San Jose, son)	12.5 hours behind India	Web app; push and SMS	Time-zoned alerts, weekly digest, the doctor-visit summary
Viewer ("Anjali", 17, granddaughter)	Wants to send voice notes and see that grandmother is fine	Web app on phone	Status only, no transcripts, send a voice note
Emergency contact ("Mr. Rao", neighbour)	No account	SMS and automated call only	Where she is, what to do, one-tap call back
Household helper ("Lakshmi", day help)	Present physically during the day	Not an app user	Pendant chimes, spoken reminders; may be recorded (consent notice)

Roles

Role	Cardinality per household	Bound to
Owner	1 or more; the last Owner cannot leave without transferring	Billing, tenant region, deletion
Caregiver	0 or more	Day-to-day operations, SOS ladder
Wearer	Exactly 1 per pendant; a household may have 2 pendants (two elders)	A pendant and a profile
Viewer	0 or more	Read status, send engagement
Emergency contact	1 or more (required to finish onboarding)	A verified phone number, no login

Permission matrix

Capability	Owner	Caregiver	Wearer	Viewer	Emergency contact
See Today status (battery, connection, last check-in)	Yes	Yes	Yes	Yes	No
Receive SOS and fall alerts	Yes	Yes	n/a	Optional	Yes (SMS + call)
Acknowledge and resolve an incident	Yes	Yes	Yes ("I'm fine")	No	Via SMS reply keyword
Create and edit reminders	Yes	Yes	Yes (voice + Simple Mode)	No	No
Read session summaries and action items	Yes	If granted "Recall access"	Yes	No	No

Capability	Owner	Caregiver	Wearer	Viewer	Emergency contact
Read full transcripts and play audio	Requires Wearer consent flag	Requires Wearer consent flag + Recall access	Yes	No	No
Start, stop, delete a recording	Yes	Yes	Yes (voice, button, app)	No	No
Change recording mode (on-request / continuous)	With Wearer consent	No	Yes	No	No
Manage devotional library and music	Yes	Yes	Yes	No	No
Send a voice note or check-in to the pendant	Yes	Yes	n/a	Yes	No
Invite members, change roles	Yes	No	No	No	No
Connect integrations (calendar, health, music)	Yes	Yes	Yes (music only)	No	No
Change AI provider and keys	Yes	No	No	No	No
Export data, change retention	Yes	No	Yes (own data)	No	No
Delete household / transfer ownership	Yes	No	No	No	No
Pendant settings (gestures, volume, fall sensitivity)	Yes	Yes	Yes	No	No
Firmware update	Yes	Yes	No	No	No

Privacy mode: who speaks for the Wearer

During onboarding the Owner chooses one of two modes, and the choice is shown on every consent screen.

- Wearer-managed (default): the Wearer accepts recording consent and transcript sharing on the pendant by voice or in Simple Mode. Family sees summaries only until the Wearer grants full transcript access.
- Family-managed: the Owner attests that they act on the Wearer's behalf (e.g. cognitive decline) and accepts consent for them. The attestation is stored in the audit log with date and name, and the Wearer is still told by voice that recording is on.

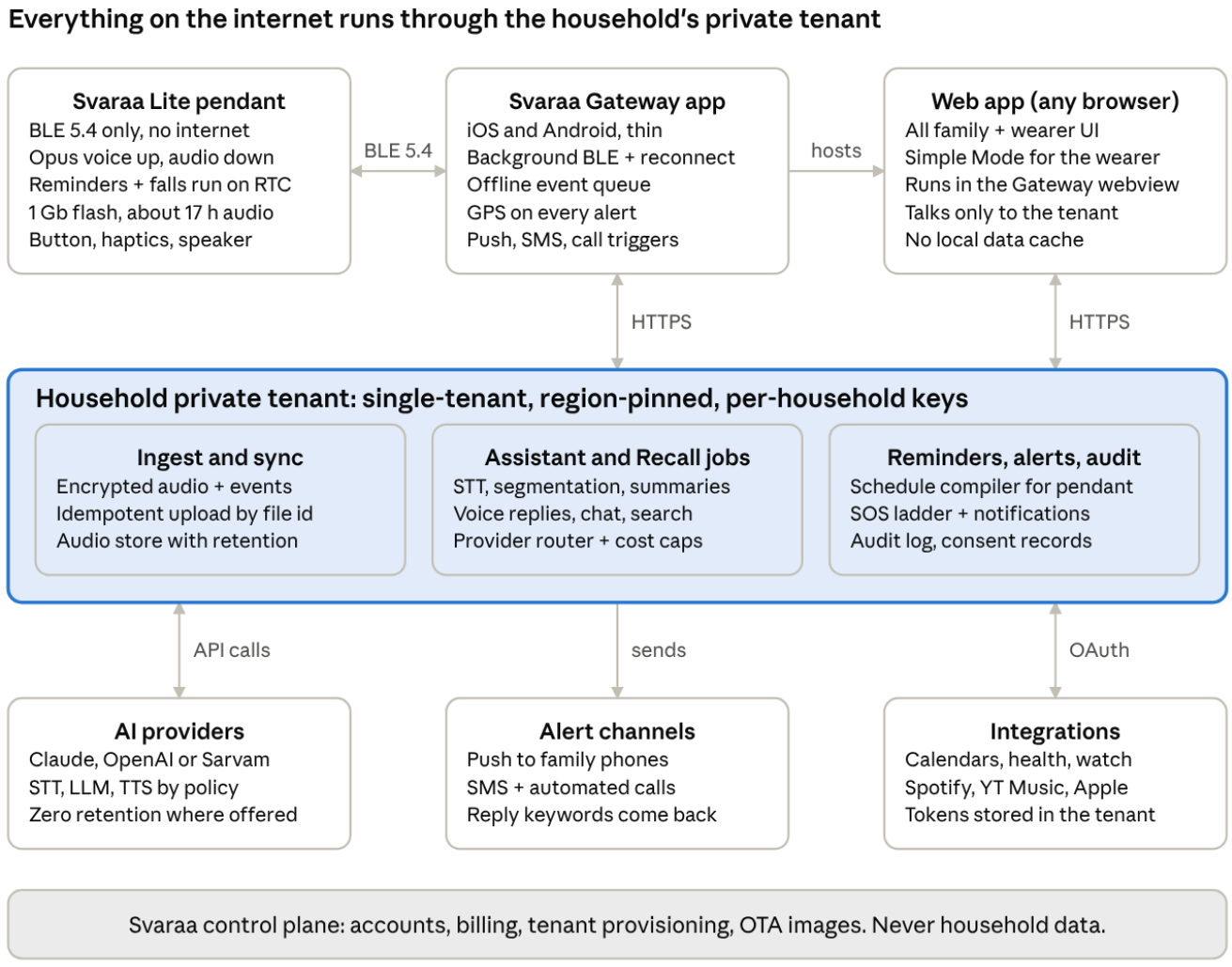
Rules and edge cases

- A person can hold roles in several households (a daughter caring for both parents in different cities); the web app shows a household switcher and never mixes alerts.
- Role changes take effect on the next request; an active SOS keeps the ladder it started with until resolved.
- Removing a member who is on the SOS ladder shows the gap and blocks the removal until the ladder still has at least two reachable contacts.
- A Wearer who is also the Owner (independent elder) sees the full app; Simple Mode is a toggle, not a restriction.
- If the sole Owner is unreachable for 90 days and billing fails, Caregivers get a 30-day notice with a "claim ownership" flow that requires two Caregivers to agree.

- A Viewer who is a minor (age entered at invite) cannot be placed on the SOS ladder and does not receive incident alerts.
- Emergency contacts are re-verified every 6 months by a test SMS; an undeliverable number flags the ladder on Today.

3. System architecture & private-tenant model

The pendant talks only to the Gateway app, the Gateway and the web app talk only to the household's private tenant, and the tenant is the only thing that ever calls an AI provider or an integration.



system architecture · pendant, gateway, web app, private tenant, providers

Read top-down: pendant events climb into the tenant through the phone, the family looks in through the browser, and everything the tenant needs from outside (AI, SMS, calendars) is a call it makes and logs itself.

Components

Component	Runs on	Responsibilities	Never does
Pendant firmware (spec §9)	nRF5340, Zephyr	Button gestures, voice uplink, reminders on RTC, fall logic, local audio files, OTA via MCUmgr	Reach the internet; delete an unsynced recording
Svaraa Gateway app	iOS / Android	BLE bonding and reconnect, sets pendant clock on every connect, pulls recordings over L2CAP with resumable offsets, encrypts audio before upload, queues events offline, supplies GPS, fires push/SMS/call, hosts the web app in a webview	Render feature UI of its own; store transcripts
Web app	Any modern browser, PWA-installable	Every screen in this document, Simple Mode, companion chat, live incident room	Hold data after logout; talk to a provider directly
Private tenant	One isolated container set per household (API, job workers, Postgres, object store) in the chosen region	Sessions, transcripts, summaries, reminders, contacts, consent records, audit log, search index, provider routing, notification routing	Share a database with another household
Control plane	Svaraa-operated, multi-tenant	Sign-up, billing, region choice, tenant provisioning and health, signed firmware images, app-store credentials for music services	Read or index any household content

Encryption and keys

- Each household has a data-encryption key (DEK) generated inside its tenant and wrapped by a region-local KMS key; the control plane never holds the DEK.
- The Gateway encrypts each recording with a per-file key derived from the household DEK before upload (spec §9 "recordings encrypted by the app before upload"); the tenant decrypts only inside the transcription job.
- Transcripts, summaries and contact details are encrypted at rest at field level; search uses an index built inside the tenant.
- Svaraa support can only reach a tenant with a time-boxed support token the Owner generates in Settings; every use is written to the household audit log.
- All links are TLS 1.3; pendant-to-phone is LE Secure Connections with bonding (spec §9).

Residency and provider egress

The tenant runs in the Owner's chosen region (India by default). AI calls may leave the region depending on the provider; the provider picker states where each provider processes audio and text, and the household's choice is stored as a consent record (section 4).

Tenant lifecycle

State	Trigger	What the family sees
Provisioning	Sign-up complete; target under 90 s	Onboarding step 2 progress card; retries silently up to 3 times, then a "we'll email you" card
Active	Health check green	Nothing

State	Trigger	What the family sees
Degraded	Job queue or storage failing	Today banner: "Summaries are delayed"; alerts still route through a control-plane fallback relay that carries only phone numbers and a location link
Suspended	Billing failed 14 days	Read-only web app; SOS, reminders and voice still work for 30 more days
Exporting	Owner requests export	Progress card; archive link valid 7 days
Destroy pending	Owner deletes household	30-day grace with restore; then keys are destroyed, which makes stored audio unreadable

Reliability rules carried over from the hardware spec

- The pendant never depends on the phone for reminders, the fall countdown or the SOS countdown (spec §9); the software must therefore treat every pendant event as possibly late.
- Events stay on the pendant until the Gateway confirms receipt, so the tenant must de-duplicate by pendant event id.
- The tenant alerts the family if the pendant has not connected for 2 h during the wearer's waking hours or was not charged for 3 days.
- OTA runs only while charging with battery at 50% or more, from signed images with rollback.

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4. AI provider layer (Claude / OpenAI / Sarvam)

The household picks one "Brain", the tenant's provider router turns that into three engines (speech-to-text, language model, text-to-speech), and no provider outage can ever stop a reminder, a fall countdown or an SOS.

Capability matrix

Model names change often; the table records the shape of each provider so the router can be configured per release. Verify the exact models, regions and retention terms at build time.

Capability	Claude (Anthropic)	OpenAI	Sarvam AI
Language model: answers, summaries, action items, chat	Yes; strongest long-context summarisation and safest refusals	Yes	Yes; Indic-tuned
Speech-to-text	None native; paired with Sarvam or OpenAI	Yes, including streaming	Yes; built for Indic and code-mixed Hindi-English
Text-to-speech	None native; paired	Yes	Yes; natural Indic voices
Indic language depth	Text strong; speech via partner	Text strong; speech good for Hindi, weaker for regional languages	Best for Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi, Gujarati, Punjabi, Odia
Processing region	US / EU per Anthropic offering	US / EU	India
Zero-retention option	Per API terms; confirm for the plan used	Per API terms; confirm for the plan used	Confirm

Routing policy

Brain chosen	Default speech-to-text	Default text-to-speech	Notes shown on the picker
Claude	Sarvam if the wearer's language is Indic, else OpenAI	Same rule	"Claude thinks, Sarvam speaks" wording; two providers are disclosed
OpenAI	OpenAI	OpenAI	Toggle: "Use Sarvam for Indian-language speech"
Sarvam	Sarvam	Sarvam	Toggle: "Send summaries to Claude" (off by default; disclosed)

Tier-0 intents (time, date, battery, next reminder, play from the library, start or stop recording, call a named contact, help) are resolved by rules inside the tenant with no provider call, so they work during a provider outage and cost nothing.

Keys and billing

- Svaraa-managed keys (default): the plan includes a monthly allowance of transcription minutes and assistant turns; usage is metered per tenant, never per Svaraa pool.
- Bring-your-own key: pasted in Settings, validated with a test call, stored encrypted in the tenant; quota and billing errors are shown to the Owner in plain words.
- Switching Brain applies to new calls only; existing summaries and transcripts are untouched. Search is provider-independent (tenant full-text index plus tenant-local embeddings), so no re-index is needed.

Fallback and cost caps

- Speech engines fail over automatically to the other speech provider on timeouts or 5xx, and the switch is logged.
- The language model fails over only if the Owner enabled "Allow fallback to another provider", because it sends text to a provider the family did not choose. With fallback off, the pendant says "I can't think right now. Your reminders still work."
- Soft cap at 80% of the monthly allowance: Today shows a usage card. Hard cap at 100%: continuous-mode transcription pauses (audio keeps recording and is transcribed when the allowance renews); on-request voice questions continue at a reduced daily limit; SOS, reminders and Tier-0 never pause.

Context and reply policy

- Every assistant call carries: wearer name and preferred languages, faith and prayer preferences, family names and relationships, today's reminder schedule, the doctor's name, current local time and city, and the three most recent session summaries when Recall access is on. Raw transcripts are added only for a question that asks about a conversation.
- Pendant replies are at most two sentences and end with a question only when a decision is needed. Web chat replies are unlimited.
- Health guardrails: the assistant never changes a dose or suggests a medication; a question about symptoms gets comfort, the family-entered doctor name and an offer to call a caregiver. Distress words trigger the check-in flow in section 11.
- Replies are in the language the wearer spoke; the wearer can switch by saying the language name.

Latency budget (single press to first spoken word)

Stage	Target p50	Target p95
Pendant to Gateway to tenant (Opus frames, streaming)	150 ms	300 ms
Speech-to-text final	500 ms	900 ms
Language model first token	600 ms	1,200 ms
Text-to-speech first audio chunk to pendant	500 ms	900 ms
Total	2.0 s	3.5 s

If no audio has started 1.5 s after the wearer stops speaking, the pendant plays a soft "thinking" tick every second so silence never reads as failure.

Edge cases

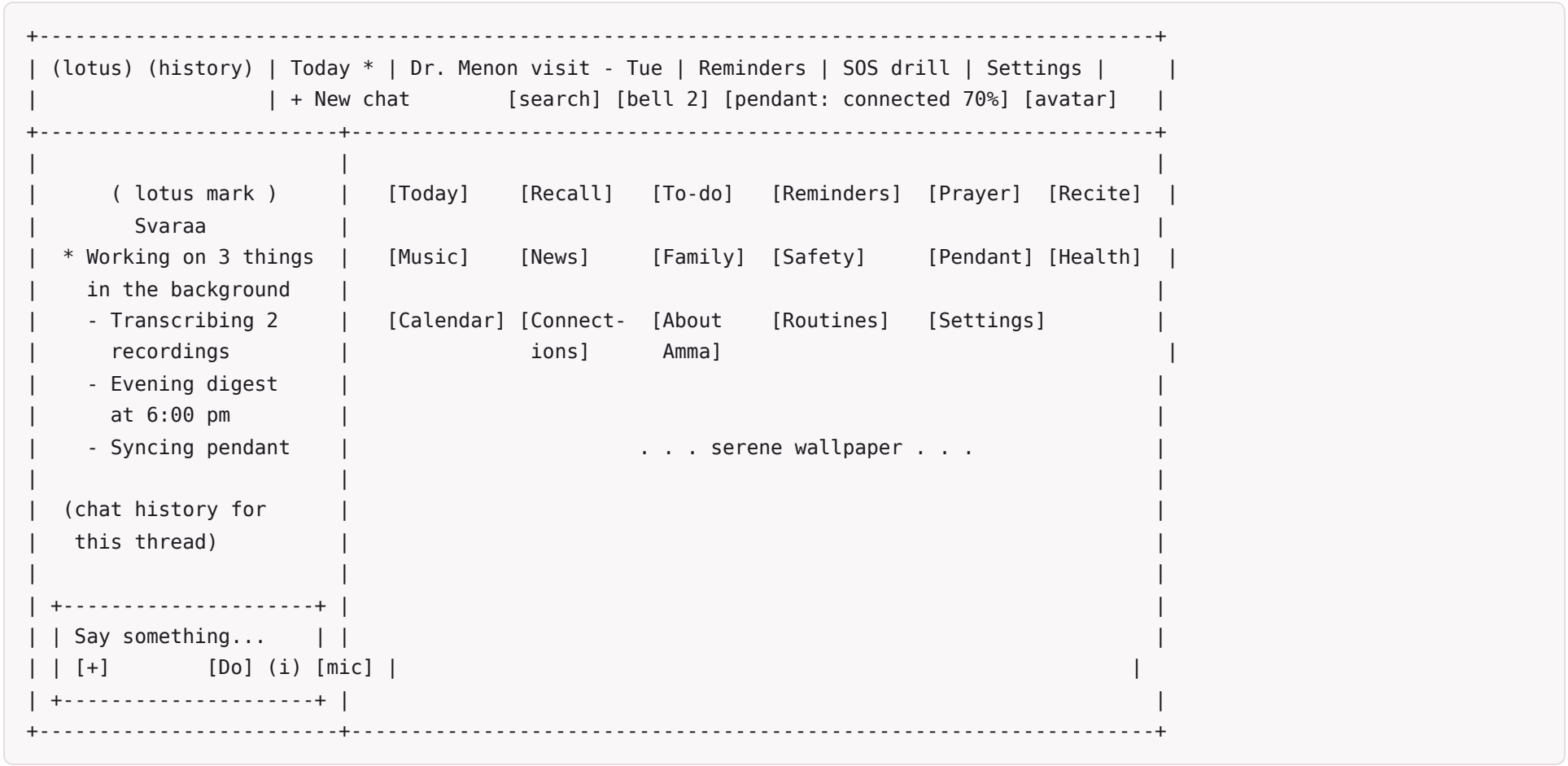
- Provider key revoked or quota exceeded: Owner gets a push and an email; the picker offers a switch to Svaraa-managed keys in one tap.
- Wrong language detected: the wearer's primary and secondary languages bound detection; a third language is transcribed but flagged for review.
- Provider refuses a request on its own content policy: the pendant says "I can't help with that one. Shall I call Priya?" and the refusal is logged without the text.
- Provider returns something that sounds like medical advice: a post-check on medication and dosage terms replaces the reply with the guardrail wording above.
- Two engines from two providers hear the same audio (Claude Brain with Sarvam speech): the disclosure is shown once at setup and again whenever the routing changes.
- Speech from visitors or the household helper reaches the provider in continuous mode: the consent notice in section 17 covers this and the wearer can mute with a triple press.

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5. UI shell: the Svaraa desktop

The web app is a calm desktop: a companion panel on the left where the family talks to Svaraa, a grid of app icons on the right over a serene wallpaper, and a top bar of tabs for whatever is open, exactly the structure of the reference screenshot.

Shell wireframe (desktop, 1280 px and up)



Shell parts

Part	Behaviour
Top bar tabs	Every app or record opened becomes a tab ("Dr. Menon visit - Tue" is one session). Tabs persist per user across devices. A dot on a tab means live activity; an incident tab pulses until resolved. Beyond 8 tabs the oldest inactive tab closes.
Pendant chip	Always visible: connection, battery, mute state, and a red "Recording" state while any recording is active. This chip is the recording indicator the pendant does not have.
Bell	Notification centre grouped into Safety, Reminders, Device, Recall, Family; Safety items pin to the top until acknowledged.
Search	Global: sessions, action items, reminders, contacts, settings. A question typed here hands off to the companion.
Companion panel	The same assistant the wearer talks to on the pendant, scoped to the viewer's role. "Working on N things" expands to the live job list (transcriptions, digests, syncs, provider fallbacks). "+" attaches a photo or file (a prescription photo proposes reminders). "Do" switches to task mode where Svaraa proposes actions as confirmation cards. The mic uses the browser microphone through the same voice pipeline.
App grid	Fixed order, drag to rearrange per user; a badge shows counts (To-do 3, Family 1). Apps a role cannot use are hidden, not greyed.
Wallpaper	Three calm scenes shipped (lake at dusk, temple morning, monsoon window); the wallpaper never carries information.

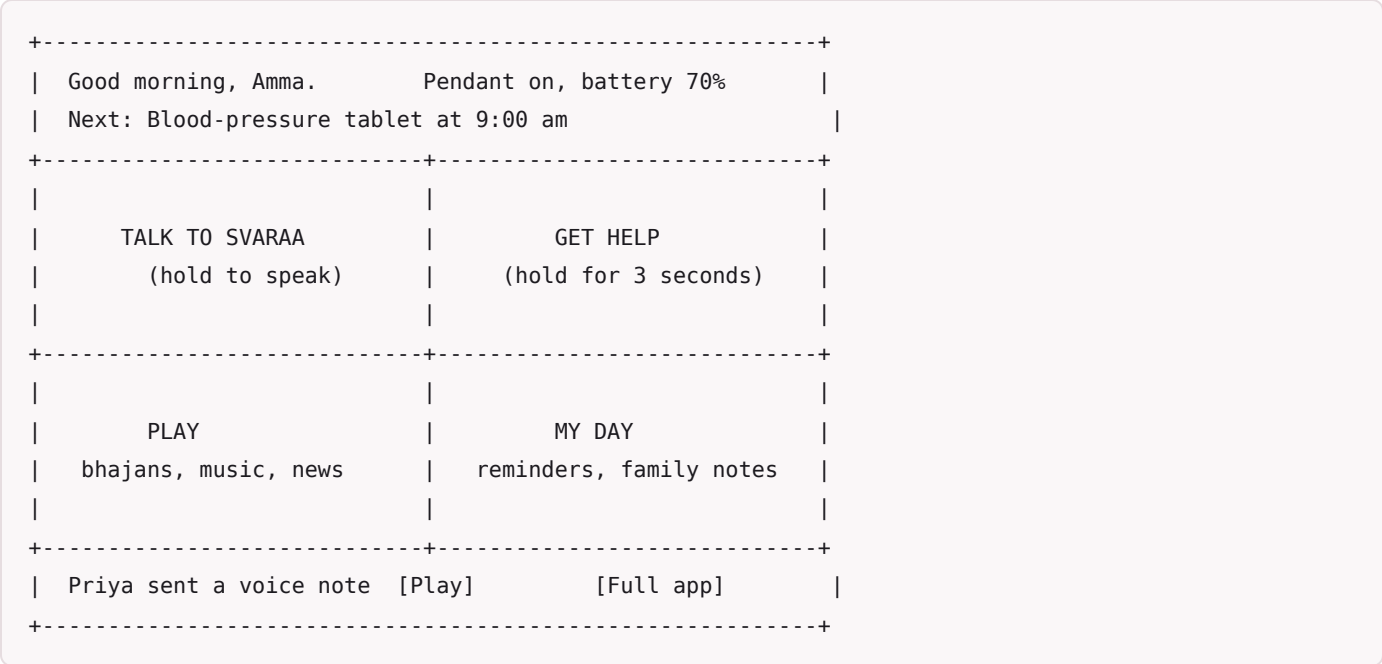
App registry

App	Opens	Roles	Section
Today	Dashboard: pendant status, timeline of the day, alerts, digest	All	7
Recall	Sessions, transcripts, summaries, search across days	Owner, Caregiver (with Recall access), Wearer	8
To-do	Action items extracted from sessions plus manual tasks, assignable to family	Owner, Caregiver, Wearer	8
Reminders	Medication, water, appointments, custom; adherence	Owner, Caregiver, Wearer	10
Prayer	Prayer-time engine, daily devotional schedule, festivals	Owner, Caregiver, Wearer	10, 12
Recite	Devotional library, playlists, play on pendant	Owner, Caregiver, Wearer	12
Music	Spotify / YouTube Music / Apple Music remote, output routing	Owner, Caregiver, Wearer	12
News	Briefing settings, topics, language, today's briefing text	Owner, Caregiver, Wearer	13
Family	Members, roles, invitations, voice notes, check-ins	All (Viewer: limited)	16
Safety	SOS ladder, fall settings, incident history, drills, live incident room	Owner, Caregiver, Wearer (read + drill)	11
Pendant	Battery, storage, gestures, volume, OTA, find my pendant	Owner, Caregiver, Wearer (subset)	15
Health	Steps, sleep, heart rate from connected sources; reminders taken	Owner, Caregiver, Wearer	14
Calendar	Connected calendars, appointments that become reminders	Owner, Caregiver, Wearer	14

App	Opens	Roles	Section
Connections	Every integration and its state	Owner, Caregiver	14
About (Wearer name)	The wearer's profile: languages, faith, doctors, medications, care notes, house map for responders	Owner, Caregiver, Wearer	6
Routines	Morning briefing, evening digest, check-in schedule, quiet hours	Owner, Caregiver, Wearer	13
Settings	Household, privacy, AI provider, billing, export, delete	Owner (full), others (own profile)	17

Simple Mode (Wearer)

Simple Mode replaces the desktop with four large tiles and a status line, with 22 px base type, no wallpaper and every control at least 64 px tall. It is a toggle on the avatar menu, remembered per device, and the Gateway app opens it by default on the wearer's own phone.



Responsive rules

- 1024 px and wider: the desktop above. 640 to 1024 px: the companion panel collapses to a rail with the "Say something" bar; the grid keeps 4 columns.
- Under 640 px (phone browser or the Gateway webview): bottom navigation with Today, Talk, Apps, Alerts; the app grid becomes a list; tabs become a "Open" sheet.
- Simple Mode is the same at every width.

Design rules

- Palette from the pendant: aubergine surfaces, rose-gold accents, cream text panels; lotus mark as the Svaraa avatar. Status colours are reserved for status (green connected, amber attention, red safety).
- WCAG AA contrast, 44 px minimum targets (64 px in Simple Mode), reduced-motion respected, screen-reader labels on every tile, Indic script font stack with Latin fallback.

- Every app has the same four states: loading skeleton, empty state with one action, error with retry, and a "pendant not connected" banner that explains what still works.
- Anything that changes the pendant (volume, gestures, a reminder) shows a "Sent to pendant" or "Waiting for pendant" chip until the pendant acknowledges.

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6. Onboarding & setup

Onboarding is a 13-step wizard that ends only when the pendant has been heard and felt, the wearer's language has been transcribed correctly, and an SOS drill has reached a real phone; it takes about 20 minutes and can be paused at any step.

Entry paths

- Family sets up: a son or daughter creates the account on a laptop, then installs the Gateway app on whichever phone will stay near the wearer.
- Wearer sets up: an independent elder runs the same wizard in Simple typography on their own phone or tablet; every text field accepts voice input.
- Add a second pendant: an existing household repeats steps 4, 6 and 8 for a second wearer.

Wizard frame

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+-------------------------------------------------------------------------------+
|  Svaraa setup                                          Save and continue later |
+------------------+--------------------------------------------+-----------------+
| 1 Account     ok |                                            |  Need a hand?   |
| 2 Private server |   Step 6 of 13: Pair the pendant           |                 |
| 3 Household   ok |                                            |  The pendant    |
| 4 About Amma  ok |   [ illustration: press the gold dot ]     |  wakes with one |
| 5 Gateway app ok |                                            |  press. If it   |
| 6 Pair pendant > |   Press the gold dot once. You will feel   |  says nothing,  |
| 7 AI Brain       |   one tick.                                |  plug in USB-C. |
| 8 Recording      |                                            |                 |
| 9 Family & SOS   |   Looking for your pendant...  (spinner)   |  [Chat with     |
| 10 Reminders     |                                            |   Svaraa]       |
| 11 Recite, music |   Found: Svaraa Lite  SL1-7F2A             |                 |
| 12 News          |   Confirm the code on the phone: 482 913   |  [Call support] |
| 13 SOS drill     |                                            |                 |
|                  |                       [Back]  [Pair now]   |                 |
+------------------+--------------------------------------------+-----------------+
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Steps

Step	Screen	Captured	Validation and logic	Skippable
1 Account	Email or phone OTP, name, country, time zone	Identity, terms consent	OTP retry with 60 s cooldown; duplicate phone offers sign-in instead	No
2 Private server	Region choice (India default), provisioning progress card	Tenant region	Provisioning target 90 s; the wizard continues to steps 3 and 4 while it runs, holding answers in memory until the tenant is ready	No
3 Household	"Who will wear the pendant?" Myself / A family member; household name; privacy mode (Wearer-managed / Family-managed with attestation)	Roles, privacy mode	Family-managed requires a typed attestation and is written to the audit log	No
4 About the wearer	Name and what Svaraa should call them, date of birth, primary and secondary language, home address and responder notes (gate code, spare key), doctor and hospital, allergies and conditions (optional, shown to responders only), faith and prayer preferences, hearing notes	Wearer profile	Address is geocoded for the responder link; prayer preferences pre-fill step 10	Partly
5 Gateway app	Which phone stays with the wearer: wearer's own / family member's / spare home-hub phone; QR or SMS link; 6-digit link code; permission walkthrough (Bluetooth, location always, notifications, Android battery exemption, iOS background modes)	Linked phone	Each permission is checked live; a missing one blocks step 6 with a fix card	No
6 Pair the pendant	Wake with one press, BLE scan, numeric-compare code, name the pendant, clock set, firmware check, hardware test (tick felt? status heard? mic level), volume	Bonded pendant	Firmware update only if charging and battery 50% or more; hardware test must pass all three checks	No
7 AI Brain	Three provider cards with where audio and text are processed; speech pairing shown; managed or bring-your-own key; live language check (wearer says one sentence, transcript shown, "Is this right?")	Provider config, disclosure consent	Language check must be confirmed; a failed check suggests Sarvam for Indic languages	No
8 Recording and privacy	Explains there is no recording light; mode choice On-request / Continuous; consent capture; transcript sharing level; retention defaults	Consent record	Wearer-managed: the wearer says "I agree" on the pendant and the audio is stored as the consent record. Wearer absent: step is deferred and becomes a Today task	Deferred only
9 Family and SOS ladder	Invite members with roles; emergency contacts without accounts; verification SMS to every contact; order tiers; channels per contact; timings	Circle, ladder	Ladder needs 2 verified contacts to finish (1 to proceed with a warning); minors cannot be on the ladder	No
10 Reminders quick start	Templates: morning medication, water every 2 h, walk, custom; prayer times auto-added from step 4; prescription photo upload proposes a schedule	Initial schedule	Every proposed reminder needs a tap to confirm; schedule is compiled and sent to the pendant	Yes

Step	Screen	Captured	Validation and logic	Skippable
11 Recite and music	Devotional playlists by tradition and language; optional Spotify / YouTube Music / Apple Music connect; output default (pendant / phone / speaker)	Library prefs, music tokens	OAuth failures leave a "connect later" card	Yes
12 News and routines	Briefing time, topics, language; evening digest time; check-in schedule; quiet hours	Routines	Quiet hours never apply to Safety	Yes
13 SOS drill	Wearer holds the gold dot 3 s; pendant counts down; alerts go out marked DRILL; screen waits for the first acknowledgement; then a simulated fall check-in ("Did you fall? Press me or say I'm fine")	Drill result	At least one acknowledgement within 5 minutes; failure opens troubleshooting (push permission, SMS route, number format)	No

Provider picker (step 7)

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+----------------------------+  +----------------------------+  +----------------------------+
|  Claude                    |  |  OpenAI                    |  |  Sarvam AI                 |
|  Thinks and summarises     |  |  Thinks, listens, speaks   |  |  Made for Indian languages |
|  Speech by: Sarvam (Tamil) |  |  Speech by: OpenAI         |  |  Speech by: Sarvam         |
|  Text processed in: US/EU  |  |  Processed in: US/EU       |  |  Processed in: India       |
|  Audio processed in: India |  |                            |  |                            |
|  Keys: Svaraa-managed  v   |  |  Keys: Svaraa-managed  v   |  |  Keys: Svaraa-managed  v   |
|          ( ) Choose        |  |          ( ) Choose        |  |          (o) Chosen        |
+----------------------------+  +----------------------------+  +----------------------------+

  Language check:  Amma, please say: "Naan inniku kaalaila nadakka ponen"   [Listening...]
  We heard: "Naan inniku kaalaila nadakka ponen"   Tamil, 98% confidence    [Yes, correct] [Try again]
```

Pairing hardware test (step 6)

```
+-----------------------------------------------------------+
|  Let's make sure the pendant works                        |
|                                                           |
|  1. Press once.       Did Amma feel one tick?   [Yes] [No] |
|  2. Press twice.      Did the pendant say its   [Yes] [No] |
|                       status aloud?                       |
|  3. Press once and    Mic level  |||||||....    [OK]       |
|     say "Hello".      We heard: "Hello"                   |
|                                                           |
|  Volume  [-]  o-------o---------  [+]   (plays a chime)   |
|                                                           |
|  Battery 62%  .  Firmware 1.0.3 (latest)  .  Clock set    |
|                                       [Back]  [Continue]  |
+-----------------------------------------------------------+
```

Completion

Today opens with a checklist card for anything skipped or deferred, a printable one-page "Amma's card" (what each press does, what each chime means, the family numbers) and a suggestion to run a second drill from a different phone.

Edge cases

- Provisioning fails three times: the wizard saves progress under the account, emails when the server is ready, and resumes at step 5.
- Pendant not found: battery may be flat from shipping; the screen says to plug in USB-C and wait for the spoken "Charging" line, then retry. Bluetooth off, or location permission missing on Android, is detected and fixed in place.
- Pendant already bonded to another phone: instructions for the service reset (hold 10 s while plugged in), then re-pair; the old Gateway gets a push that its pendant was moved.
- Firmware update interrupted: MCUboot dual-slot rollback; the wizard shows "still on 1.0.2, will retry tonight while charging".
- iOS after a reboot needs the Gateway launched once before background BLE resumes; the wizard and Today both show this as a fix card when the phone reports a fresh boot.
- Gateway installed on a family member's phone that leaves the house: the wizard warns that the phone must stay within 10 to 20 m of the pendant and offers the spare home-hub phone option.
- Verification SMS undelivered (carrier filtering, DND registries): fall back to a voice-call verification that reads a 4-digit code.
- Wearer declines recording consent: Recall and Repository are switched off, every other feature works, and Today shows "Recording is off by Amma's choice".
- Wearer's language unsupported by the chosen speech engine: the picker blocks the choice and recommends the provider that supports it.

- Drill gets no acknowledgement: troubleshooting checks push permission on each phone, SMS route health and number formats, then offers a retry; the household cannot leave the wizard with an unverified ladder.
- Wizard abandoned: a reminder email at 24 h and 72 h; a paired pendant with no ladder routes SOS to the Owner's own phone until the ladder exists.
- Owner in one time zone, wearer in another: every schedule is stored in the wearer's zone; the Owner's screens show both times.
- Two people run the wizard at once: the second gets a "Priya is already setting up" notice and read-only steps.

---

7. Today dashboard & companion panel

Today answers the family's one question, "is Amma alright and what needs me?", in a single screen: a status strip, an attention list, and a live timeline of the wearer's day that fills as the pendant syncs.

Today wireframe

```
| Today  .  Saturday 26 Sep  .  Amma's household                [Digest at 6:00 pm] [Print Amma's card] |
+-----------------------------------------------------------------------------------------------------+
| Pendant: connected, 70%  |  Phone: Amma's phone, at home, seen 2 min ago  |  Last spoke: 12 min ago  |
| Recording: off           |  Next: Evening aarti 6:30 pm                   |  Brain: Sarvam           |
+----------------------------------------------+------------------------------------------------------+
| Needs attention (2)                          | Timeline (live)                                      |
|  ! 1:00 pm Blood-pressure tablet not         | 06:30  Morning aarti played on the pendant (28 min)  |
|    acknowledged (40 min ago)                 | 07:05  Morning briefing: 3 news items, weather       |
|    [Call Amma] [Mark taken] [Snooze 30 min]  | 08:10  Session: Phone call with Ravi (14 min)        |
|  ! Pendant not charged for 3 days            |        "Ravi visits on 12 Oct; bring the scan report"|
|    [Send a charge reminder to the pendant]   |        2 action items                    [Open]      |
|                                              | 09:00  Reminder: BP tablet, taken (one press)        |
| Action items due (3)                         | 11:20  Voice question: "What time is aarti tonight?" |
|  [ ] Book cab for Dr. Menon on Tue    Priya  | 12:40  Session: Visit from Mrs. Rao (22 min)  [Open] |
|  [ ] Buy Amlodipine refill            Ravi   | 13:00  Reminder: BP tablet, MISSED                   |
|  [ ] Send Ravi the scan report        Amma   | 13:10  Family: Anjali's voice note played            |
|                                              | 14:00  Check-in: "How are you feeling?" - "Fine"     |
| Family today                                 | 14:32  Pendant synced 2 recordings (transcribing)    |
|  Anjali sent a voice note                    |                                                      |
|  Ravi read the Dr. Menon summary             | [Yesterday]  [This week]                             |
+----------------------------------------------+------------------------------------------------------+
```

What each part shows

Part	Source	Rules
Status strip	Gateway heartbeat every 60 s; pendant battery and mute state from BLE events; last spoke from the last voice uplink	"Seen N min ago" turns amber at 30 min and red at 2 h during waking hours (the spec's reliability rule); Recording shows red while any recording is active
Needs attention	Missed reminders past their escalation window, device rules (not charged 3 days, battery under 20%, storage nearly full), deferred consent, overdue drill (90 days), failed integrations	Each item carries one primary action; items clear when the underlying state clears, never by dismissal alone
Action items due	To-do app, filtered to today and overdue	Assignee chips; checking one writes back to the session it came from
Family today	Audit log entries the viewer is allowed to see	A Viewer sees only their own entries
Timeline	Reminders, sessions, voice questions (question text only, no answer), recitations, check-ins, voice notes, device events, incidents	Session rows show a one-line summary and open the session tab; a Viewer sees "Conversation, 22 min" with no text; incidents pin to the top of the timeline until resolved

Evening digest

At the household's digest time (default 6:00 pm wearer time) the tenant compiles the day: sessions and their takeaways, new action items, reminders taken and missed, recitations played, check-in answers, steps if Health is connected, and device notes. It is posted to Today, pushed to Owner and Caregivers, and read aloud to the wearer on request ("Svaraa, how was my day?"). The digest never states mood or health conclusions; it reports what happened.

Companion panel

The panel is the same assistant the wearer hears, scoped to the viewer's role. Its status line lists live jobs; each job opens the tab it belongs to.

```
|  * Working on 3 things                          |
|    - Transcribing 2 recordings (Sarvam)  2 min  |
|    - Evening digest scheduled 6:00 pm           |
|    - Syncing pendant: 1 of 2 files, 64%         |
|                                                 |
|  Priya: Move Amma's BP tablet to 8:30 and       |
|         tell Ravi                                |
|  Svaraa: Here is what I will do:                |
|    [x] Change "Blood-pressure tablet" from       |
|        9:00 to 8:30 every day                   |
|    [x] Send Ravi: "Amma's BP tablet is now       |
|        at 8:30"                                 |
|    Note: the pendant will confirm the new time  |
|    when it next connects.                       |
|                     [Do it]  [Edit]  [Cancel]   |
```

- Ask mode answers from household data ("When is the next appointment?", "What did Dr. Menon say about salt?") with links to the source session or reminder.
- Do mode proposes actions as a checklist card; nothing changes until "Do it". Actions available: create, move or delete reminders; assign action items; send a voice note or text to the pendant; start a check-in; play something on the pendant; open a session; draft a message to a family member.
- Attach (+) accepts a prescription or discharge-summary photo and returns proposed reminders and profile notes for confirmation.
- Threads: "New chat" starts a thread inside the panel; threads are per user and searchable.

Realtime and freshness

The web app holds a server-sent event stream from the tenant; on loss it polls every 30 s and shows "Updating..." in the status strip. Timeline rows carry the pendant timestamp, so a late-synced event appears at its true time with a "synced 2 h later" note.

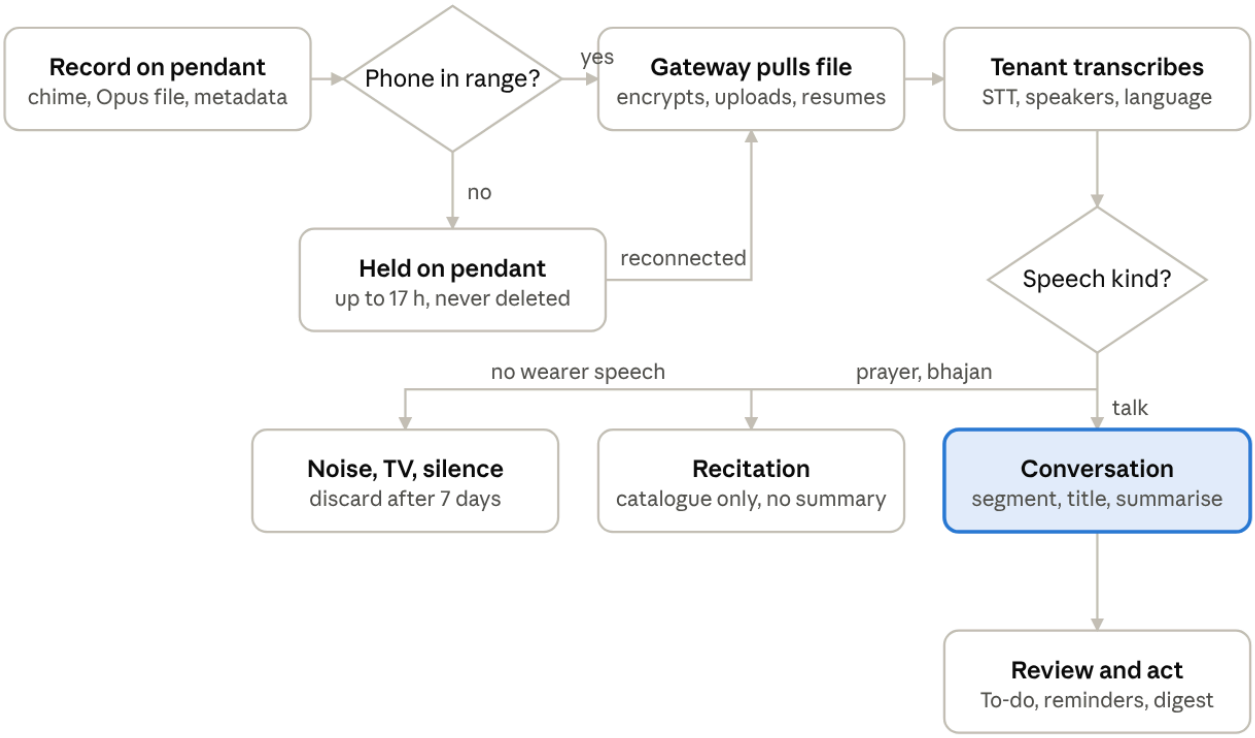
Edge cases

- Pendant offline: the strip turns amber, the attention list adds "Pendant not connected since 1:10 pm", and every action that needs the pendant shows "Waiting for pendant" instead of failing.
- Nothing has happened yet today: the timeline shows the day's planned reminders and routines in grey instead of an empty state.
- Two wearers in the household: Today gets a wearer switcher in the title; alerts stay per wearer.
- Tenant degraded: a banner says which parts are delayed (summaries, digest) and that safety alerts are unaffected.
- Wearer opens Today in Simple Mode: it shows the status line and the next three items only, read aloud on tap.
- Late events reorder the timeline: the newest-synced item briefly highlights so the reader notices it arrived.
- Digest time falls inside an active incident: the digest is delayed until the incident is resolved and states that it was.

8. Recall & Repository: recordings, sessions, summaries

Every recording the pendant makes is pulled by the phone, encrypted, transcribed on the household tenant, cut into sessions by silence, place and people, and summarised into takeaways and action items that land in To-do and the evening digest.

A recording becomes a summarised session only after a speech-kind check



recording pipeline · pendant to summarised session

A file that never reaches the phone waits on the pendant; a file that reaches the tenant is transcribed, sorted by what kind of speech it holds, and only real conversation is summarised.

Recording modes

Mode	How it starts and stops	Pendant behaviour (spec §7)	Default
On-request	Single press then "record this", the web app, or a voice command; stops on "stop recording", a press, or 60 minutes	Rising chime at start, falling chime at stop, soft chime every 10 minutes	Yes
Continuous	Enabled by the wearer (or Owner in Family-managed mode) after the consent record; runs during waking hours as rolling 10-minute files	Same chimes; triple press cuts microphone power and says "Microphones are off"	Off
Scheduled	A calendar event tagged "record" (doctor visit) starts and stops recording at its bounds	Spoken "Recording your appointment" at start	Off

Session segmentation rules

1. Units: each on-request file is a candidate session; continuous-mode files are stitched into one stream per day before cutting.
2. Hard boundaries: a silence gap of 4 minutes or more (household setting 2 to 10), a location change of more than 200 m from the phone's GPS, the start or end of a calendar event, and a detected phone call on the Gateway phone.
3. Soft boundaries: a sustained change in the set of speakers for more than 2 minutes, scored together with topic shift by the language model; a boundary is cut only when both agree.
4. Merges: two on-request files less than 90 s apart become one session (accidental double start).
5. Length cap: a session longer than 90 minutes is split at its longest pause.
6. Speech-kind check per segment: conversation, recitation (matches what the pendant was playing, or prayer-language detection), or no wearer speech (TV, radio, silence). The wearer's voice is recognised from the one-sentence language check recorded in onboarding.
7. Titles: at most six words, naming the known participant ("Phone call with Ravi", "Visit from Mrs. Rao", "Dr. Menon: BP review"); the wearer can rename by voice.
8. Sessions belong to the day they started; a session crossing midnight stays on the earlier day.
9. Human edits (split, merge, rename, delete, private) are pinned and survive any reprocessing.

Summary schema

Field	Content	Rule
Takeaways	Up to five sentences of what was said or decided	Each links to the transcript lines it came from
Action items	What, who (suggested from family names), by when	Created in To-do as "proposed" until someone confirms
Mentions	People, medicines, dates, places, amounts	A medicine or a date shows an "Add reminder" or "Add to calendar" button; never applied automatically
Follow-ups	Questions worth asking next time	Shown in the session only
Tone	None	The summary reports; it does not judge mood or health

Recall app wireframe (day view)

```
| Recall  .  Sessions                                  [Search: "what did the doctor say about salt"] |
+------------+-------------------------------------------------------------------------------------+
| Sep 2026   |  Saturday 26 Sep                                          Filter: [All] [Calls] [Visits]  |
|  26 Sat  3 |                                                                                     |
|  25 Fri  1 |  08:10  Phone call with Ravi                               14 min  .  2 action items       |
|  24 Thu  0 |         Ravi visits on 12 Oct; bring the scan report to Dr. Menon; refill Amlodipine       |
|  23 Wed  2 |         [Open]  [Play]  [Share with family]                                                 |
|  22 Tue  1 |                                                                                     |
|  ...       |  12:40  Visit from Mrs. Rao                                 22 min  .  1 action item        |
|            |         Temple committee meeting on Sunday; Mrs. Rao's son is back from Dubai              |
| Unsorted   |         [Open]  [Play]  [Mark private]                                                      |
| audio (1)  |                                                                                     |
|            |  15:05  Recitation: Hanuman Chalisa (played from library)  12 min  .  catalogued            |
| Storage    |                                                                                     |
| 3.2 GB     |  Waiting: 1 recording syncing from pendant (64%)                                            |
+------------+-------------------------------------------------------------------------------------+
```

Session detail wireframe

```
| Session: Phone call with Ravi   Sat 26 Sep, 08:10 to 08:24  .  Home  .  Amma, Ravi     [Rename] [...]  |
+-------------------------------------------------------------------------------------------------------+
|  [Summary]   [Transcript]   [Audio]                                          Ask about this session:  |
|                                                                              [ When is Ravi coming? ] |
|  Takeaways                                                                                            |
|   . Ravi arrives on 12 Oct for two weeks and will stay at home.                      (lines 4-9)      |
|   . Amma should bring the scan report to Dr. Menon on Tuesday.                      (lines 21-23)     |
|   . The Amlodipine strip runs out this week.                                        (lines 30-31)     |
|                                                                                                       |
|  Action items                                                                                         |
|   [ ] Bring scan report to Dr. Menon       Amma    Tue 29 Sep     [Confirm] [Assign v] [Dismiss]      |
|   [ ] Buy Amlodipine refill                Ravi    this week      [Confirm] [Assign v] [Dismiss]      |
|                                                                                                       |
|  Mentions                                                                                             |
|   Amlodipine  [Add reminder]    12 Oct  [Add to calendar]    Dr. Menon  [Open contact]                |
|                                                                                                       |
|  Transcript (tab)   08:10:04  Ravi   Amma, how are you feeling today?                  [play]         |
|                     08:10:09  Amma   Fine, fine. Did you book the ticket?              [play]         |
|                     ...                                                                               |
|  Tools: [Split here] [Merge with previous] [Mark private] [Delete] [Export PDF]                       |
+-------------------------------------------------------------------------------------------------------+
```

To-do app wireframe

Action items from every session land here as "proposed" until a person confirms them; manual tasks and family assignments live alongside, and each item keeps its link to the line it came from.

```
| To-do  .  Amma's household                        [All] [Proposed 2] [Mine] [Amma's]        [+ Add task] |
+-------------------------------------------------------------------------------------------------------+
|  Proposed from sessions                                                                               |
|   [ ] Bring scan report to Dr. Menon     from: Phone call with Ravi (Sat)   Amma   Tue   [Confirm] [Dismiss]|
|   [ ] Buy Amlodipine refill              from: Phone call with Ravi (Sat)   Ravi   this week [Confirm]     |
|  Today                                                                                                |
|   [ ] Book cab for Dr. Menon             Priya     Mon        added by Priya                            |
|   [ ] Send Ravi the scan report          Amma      today      Svaraa will remind Amma at 5 pm  [Edit]      |
|  This week                                                                                            |
|   [ ] Temple committee: confirm Sunday   Amma      Sat        from: Visit from Mrs. Rao                 |
|  Done (3)                                                                                             |
+-------------------------------------------------------------------------------------------------------+
```

- A task assigned to the wearer can carry a spoken reminder; a task assigned to a family member sends a push and appears in their own To-do across households.
- Completing a task writes back to its session; dismissing a proposal teaches the summariser nothing about the wearer, only that the item is closed.

Search and voice recall

- Web search combines full-text and tenant-local semantic search, filtered by date, participant and kind; results show the passage and open the session at that line.
- Voice: "What did the doctor say on Tuesday?" is resolved by date and participant first, then the top passages are summarised into two spoken sentences, ending with "Want the full summary on your phone?".
- "Svaraa, delete that" removes the most recent recording and its transcript after a spoken confirmation; "forget the last 10 minutes" trims a continuous stream; "pause listening for an hour" suspends continuous mode.

Storage and retention

- Raw audio is kept 30 days by default (options 7, 30, 90, keep); transcripts and summaries stay until deleted; any session can be pinned to keep its audio.
- The pendant deletes its local copy only after the tenant confirms the file is stored (spec §9 list, pull, delete over L2CAP).
- Storage used, and the cost of continuous mode in transcription minutes, are shown in the Recall rail and in Settings.

Edge cases

- Partial file because the battery died mid-recording: transcribed as far as it goes, marked "cut short".
- Recorded before the clock was set (RTC reset after a flat battery): timestamped from sync time with an "approximate time" flag; the Gateway sets the clock on every connect.
- Duplicate upload after a dropped connection: de-duplicated by pendant file id; resumable by byte offset.
- Pendant storage above 80%: the pendant says "My memory is nearly full" and Today shows an attention item; at 100% new recordings refuse with a double falling chime.
- Large backlog after a day out of range: oldest files first, voice-question audio prioritised; the Recall rail shows the queue and an estimate.
- Transcription fails: three retries, then failover to the other speech engine if allowed, then a "needs attention" item with a manual re-run.
- Mixed languages in one session: transcribed as spoken; the summary is written in the household's reading language.
- Someone else pressed record and the wearer never spoke: classified as "no wearer speech", kept 7 days, never summarised.
- Visitors and household help are recorded: the consent notice in section 17 applies; the wearer can mark a session private, which hides it from everyone but the wearer.
- Summary names the wrong person: every takeaway links to its lines, and a corrected name is remembered as a speaker label for future sessions.
- Household hits the transcription cap: continuous-mode audio is stored and queued; on-request sessions still transcribe under the reduced limit.

- Gateway phone replaced: files in transit are re-pulled from the pendant by the new phone; nothing is lost because the pendant keeps files until confirmed.
- Wearer travels across time zones: sessions keep the pendant's local time and show the zone.

---

9. Respond: the voice assistant

One press, one question, one short answer: the assistant is tuned for a wearer who cannot see a screen, so it answers in two sentences, in the language spoken, and turns anything that sounds like distress into a check-in.

Interaction loop on the pendant

1. Single press: one tick; the pendant streams microphone audio to the Gateway and on to the tenant.
2. End of speech: 1.2 s of silence, a second press, or a 30 s cap; the tenant returns a streamed spoken reply.
3. A press during the reply stops it. If the reply ends with a question, the microphone reopens for 6 s with a tick so the wearer can answer without pressing again.
4. A short-term memory of the last 10 turns within 30 minutes makes "and tomorrow?" work.
5. Facts the wearer states about themselves ("my grandson is Arjun") are proposed as profile notes; in Wearer-managed mode they save on a spoken "yes", otherwise a Caregiver confirms.

Intent catalogue

Intent	Examples	Tier	Handling	Reply pattern
Status	"Battery?", "Are you connected?"	0	Pendant answers locally, same as double press	"Battery seventy percent, connected to your phone."
Time, date, day, festival	"What day is it?", "When is Diwali?"	0 / 1	Rules for time; festival calendar from Prayer settings	One sentence
Next reminder, adherence	"Did I take my tablet?", "What's next?"	0	Reminder engine state	"You took the blood-pressure tablet at nine. Next is water at eleven."
Create or change a reminder	"Remind me to call Ravi at six"	1	Parsed to a proposal, read back, saved on "yes"; pendant schedule updated	Read-back then confirmation
Record, stop, delete, pause	"Record this", "Stop", "Delete that", "Pause listening for an hour"	0	Section 8	Chime plus one word
Play devotional	"Play Hanuman Chalisa", "Morning aarti"	0	Library lookup, stream to the pendant	Starts playing; "Playing Hanuman Chalisa"
Play music	"Play old Tamil songs on Spotify"	1	Music remote (section 12) on the phone or speaker	"Playing on your phone" or the reason it cannot
Volume, stop playback	"Louder", "Stop"	0	Pendant or phone volume	Chime
Call a family member	"Call Priya"	0	Android: Gateway dials on speaker. iOS: call sheet on the phone plus a push to Priya "Amma wants to talk"	"Calling Priya" or "I've asked Priya to call you"

Intent	Examples	Tier	Handling	Reply pattern
Send a note	"Tell Priya I'm going to the temple"	1	Transcribed, sent as push and shown in Family	"Sent to Priya"
Recall	"What did the doctor say on Tuesday?"	1	Section 8 search over sessions	Two sentences plus an offer of the full summary
Prayer times	"When is Maghrib?", "Aarti time?"	0	Prayer engine (section 10)	One sentence
News, weather	"What's the news?"	1	Briefing engine (section 13)	Headlines, then "Want more on any of these?"
General questions, stories, scripture, translation	"Tell me a story", "What does this shloka mean?"	1	Language model with the context policy	Two sentences unless the wearer asks for more
Wellbeing words	"I'm not feeling well", "I'm lonely", "I fell yesterday"	1	Comfort, offer to call a caregiver, log a wellbeing note; a fall mention notifies Caregivers as a note, not an SOS	Warm, one question
Help by voice	"Help me", "Bachao", "Kaapathu"	0	Starts the SOS countdown (section 11); on by default, can be disabled	Countdown speech
Unclear	Low-confidence transcript	0	Ask once, then stop	"Could you say that again?" then "I didn't catch that. Press me and try again."

Voice persona

- Warm and brief; addresses the wearer by the chosen name with the chosen honorific; never lists, never reads URLs, speaks numbers as words and dates relatively ("this Tuesday").
- Answers the same question patiently every time; repetition is never pointed out to the wearer. A count of repeated questions appears in the digest only if the family opts in to "repetition counts".
- Health and medication guardrails from section 4 apply to every reply.

Web chat

The companion panel (section 7) uses the same intent catalogue plus family-only actions. Voice questions from the pendant appear in a "Voice log" inside Recall: the wearer sees questions and answers; family with Recall access sees the questions only; answers are kept 30 days for quality review then dropped.

Wake word (V1.1, designed now)

A "Hey Svaraa" toggle in Pendant settings ships disabled; enabling it shows the battery cost from the hardware spec (about 155 mAh per day in the heavy profile) and the licence notice.

Edge cases

- Phone out of range: the pendant itself says "I can't reach your phone" with two slow pulses (spec §7); nothing is queued because a question has no value later.
- Phone in range but no internet: the Gateway answers Tier-0 intents from its cached schedule and says "I can't think right now; your reminders still work" for the rest.

- Accidental press: 3 s of silence closes the microphone with one tick and no speech.
- Noisy room: low-confidence transcript triggers one "say that again"; the second failure ends the turn.
- Wearer keeps talking past 30 s: the assistant answers what it heard so far and says it is listening again.
- Two bonded pendants near the same phone: each has its own identity and profile; audio streams are never mixed.
- Wearer speaks a third language: transcribed best-effort and answered in the primary language.
- Casual mention of a fall or pain: logged as a wellbeing note and pushed to Caregivers with the exact words; never escalated to SOS without the wearer's confirmation.
- Provider refusal or outage: section 4 fallback wording.
- Late-night questions: answered normally; quiet hours only affect notifications, never the wearer's own requests.

---

10. Remind: reminders, medication & prayer times

Reminders fire from the pendant's own clock, so they work with the phone in another room; the software compiles a seven-day schedule the pendant can hold, watches for acknowledgements, and tells the family when a medication goes unanswered.

Reminder types

Type	Set by	Spoken on pendant	Acknowledge	Escalates to family
Medication	Family, wearer, prescription photo	"Time for your blood-pressure tablet, after breakfast"	Single press = taken; "later" = snooze 15 min	After 30 min unacknowledged (default; per reminder)
Water and meals	Interval or times	"Time to drink some water"	Press or ignore	Never
Prayer	Prayer engine (below)	"It is time for evening aarti"; can auto-play a recitation	Press or ignore	Never
Appointment	Calendar or manual	"Dr. Menon at 10 tomorrow; the cab comes at 9:15" spoken the evening before and 1 h before	Press	If the wearer is not at the location by start time and location is shared (optional)
Check-in	Routines	"How are you feeling today?" waits for a spoken answer	Voice answer	If unanswered twice in a day
Charge nudge	System	"Please charge me soon" at 15% (firmware); the app repeats at 20% by voice note	None	Family at 20% (spec §6)
Custom	Voice or web	Any text	Press	Optional

Reminders app wireframe

```
| Reminders  .  Amma                              [+ Add]  [Prescription photo]  [Adherence]  [Pendant: in sync] |
+-------------------------------------------------------------------------------------------------------------+
| Morning                                                                                                      |
|  06:00  Morning prayer (Prayer engine)     daily        plays: Suprabhatam 12 min     [Edit]                 |
|  09:00  Blood-pressure tablet, Amlodipine  daily   *important*   escalate 30 min       today: taken 09:02    |
|  09:30  Water                              every 2 h until 8 pm                        today: 3 of 6         |
| Afternoon                                                                                                    |
|  13:00  Blood-pressure tablet              daily   *important*                          today: MISSED  [Call] |
|  14:00  Check-in: "How are you feeling?"   daily                                       today: "Fine"         |
| Evening                                                                                                      |
|  18:24  Evening aarti (sunset-based)       daily        plays: Om Jai Jagdish 6 min                          |
|  20:00  Ravi's weekly call                 Sundays                                                           |
| Tomorrow                                                                                                     |
|  09:15  Cab to Dr. Menon (from Calendar)   once        spoken tonight at 8 and at 8:15 tomorrow              |
+-------------------------------------------------------------------------------------------------------------+
| Pendant holds 41 of 64 slots for the next 7 days  .  Last confirmed by pendant 14:32                         |
+-------------------------------------------------------------------------------------------------------------+
```

Add or edit reminder

```
+------------------------------------------------------------+
|  New reminder                                              |
|  What Svaraa says   [ Time for your blood-pressure tablet ]    |
|  Language           [ Tamil v ]      Preview: [Play on pendant]|
|  When               [ 09:00 ] [ 13:00 ] [+ time]               |
|  Repeat             (o) Daily  ( ) Weekdays  ( ) Custom        |
|  Type               [ Medication v ]   Medicine [ Amlodipine ] |
|  Important          [x] repeat every 5 min until acknowledged  |
|                     [x] tell family after [30] min             |
|  Quiet hours        [ ] may fire during quiet hours            |
|  Also play          [ none v ]                                 |
|                                    [Cancel]  [Save to pendant]|
+------------------------------------------------------------+
```

Schedule compiler

- The tenant expands every recurrence 7 days ahead into occurrences, sorts by priority (important medication, appointments, prayer, check-ins, water, custom) and writes the first 64 into the pendant's compact schedule; the slot count is a firmware constant to confirm in EVT.
- The schedule is pushed whenever its hash differs, on every connect; the pendant acknowledges, and the app shows "in sync" or "waiting for pendant" with the time of the last confirmation.
- The Gateway caches the same schedule so the wearer's phone can speak a reminder if the pendant is off the wrist and charging.
- The pendant repeats an unacknowledged reminder after 5 minutes (spec §7) and reports acknowledgement events with its own timestamp.

Prayer-time engine

Tradition	Computed	Set by the family	Extras
Hindu	Sunrise and sunset for the home location (aarti, sandhya)	Puja times, deity-specific days, weekly fasts	Regional festival calendar (panchang data source to select, section 20); Ekadashi and fasting reminders
Islam	Fajr, Dhuhr, Asr, Maghrib, Isha from location with the chosen calculation method (MWL, Karachi, ISNA, Umm al-Qura) and Asr juristic setting; minute offsets	Jumu'ah time	Ramadan mode: suhoor and iftar reminders, later quiet hours
Christian	Sunrise and sunset options	Morning and evening prayer, Sunday service	Liturgical calendar dates
Sikh	Amrit vela (before sunrise), Rehras at sunset, Kirtan Sohila at night	Gurdwara visits	Gurpurab dates
Custom	None	Any times	Any recitation attached

Times recompute every night for the next day from the phone's last known location; a location change of more than 50 km recomputes immediately and tells the wearer once ("Prayer times updated for Bengaluru").

Adherence view

A week grid per medication (taken, late, missed, unknown) with the time acknowledged; "unknown" means the pendant was offline at fire time. The weekly digest includes the counts; nothing in the app suggests doubling or skipping a dose.

Escalation ladder for an important reminder

1. T+0: pendant speaks and vibrates three pulses.
2. T+5 min: pendant repeats (firmware). T+10: repeats again.
3. T+15 min: if the wearer's own phone runs the Gateway, it also speaks the reminder.
4. T+30 min (configurable): push to Caregivers and Owner: "Amma hasn't acknowledged the blood-pressure tablet" with Call, Mark taken, Snooze 30 min.
5. Any acknowledgement, from the pendant or the app, stops the ladder and is logged with who and how.

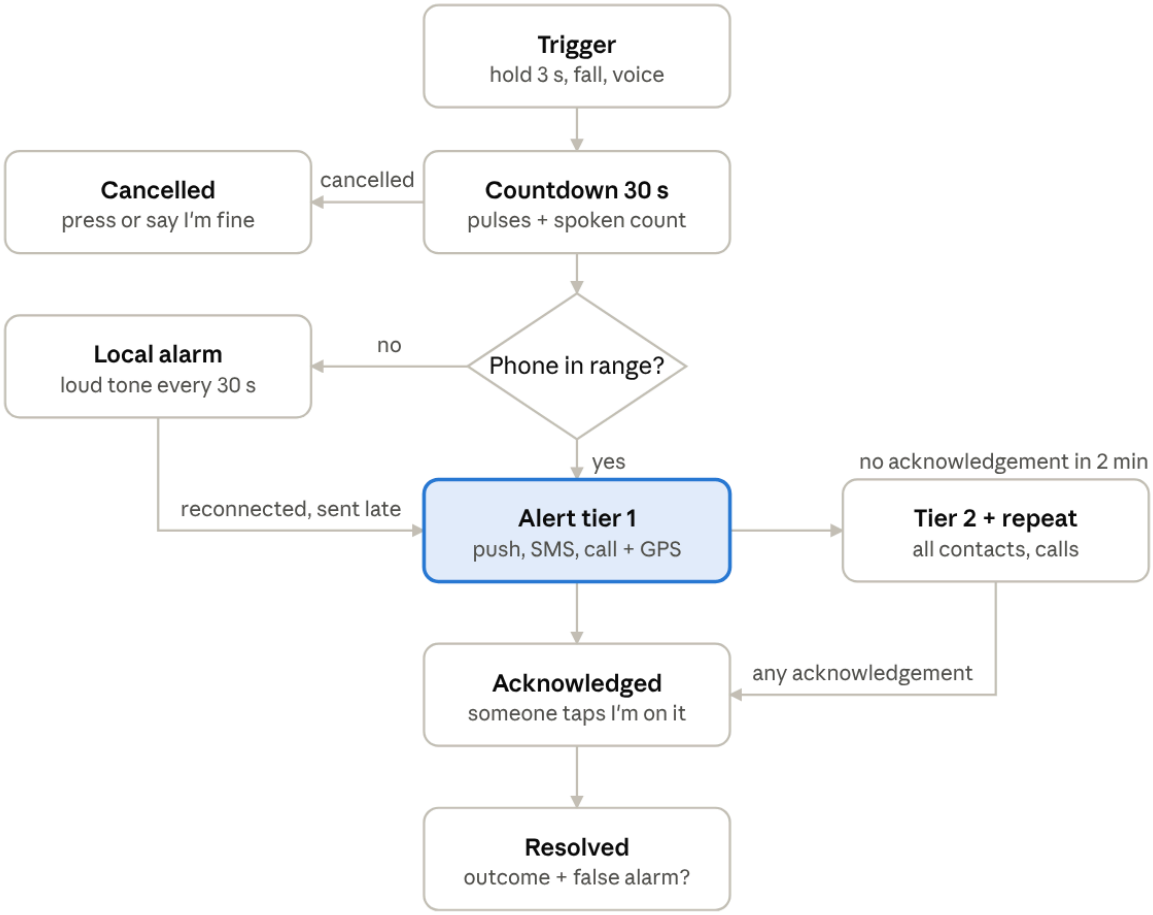
Edge cases

- Pendant offline at fire time: the reminder still fires on the pendant; the app shows "unknown" until the event syncs, then backfills taken or missed with the pendant time.
- Pendant not connected for 7 days: the schedule on it is stale; Today warns that reminders beyond the cached week will not fire.
- Single press is both "talk" and "acknowledge": during a reminder's 5-minute window a press acknowledges; outside it, a press listens.
- Two reminders within 2 minutes: spoken as one sentence ("Time for your tablet and a glass of water"); one press acknowledges both.
- Wearer says "I already took it" before the time: marked taken early; the scheduled occurrence is skipped and logged.
- Medication changed mid-day: the compiler re-issues the schedule; occurrences already fired are untouched.
- Prescription photo misread: every proposed line needs a tap; unreadable lines are left blank, never guessed.
- Calendar appointment moved or cancelled: the linked reminders move or delete and Today notes it.
- Quiet hours (default 10 pm to 6 am): only reminders flagged "may fire during quiet hours" speak; others vibrate once and are re-spoken at the end of quiet hours.
- Triple-press mute: microphones are off but the speaker and haptics work, so reminders still fire; the app shows the mute state.
- Battery at 5%: the pendant keeps only the button, fall alerts and reminders (spec §6), so reminders survive; recitations attached to them do not play.
- Owner in a different time zone: all times are entered and shown in the wearer's zone with the Owner's local time in grey.
- Hearing-impaired wearer: a per-wearer setting makes reminders vibration-first with the phone speaking the text.

11. Rescue: SOS & fall management

An incident starts on the pendant, counts down where the wearer can cancel it, and then climbs a ladder of family contacts until a person says "I'm on it"; if the phone is out of range the pendant raises a local alarm and the alert goes out the moment it reconnects.

An SOS always reaches a person, even when the phone is out of range



SOS and fall incident lifecycle · trigger to resolution

The countdown belongs to the pendant, the ladder belongs to the tenant, and a late-synced event still produces a real alert marked with how long ago it happened.

Triggers

Trigger	Detection	Countdown wording	On by default
Hold the gold dot 3 s	Firmware	"Calling for help in thirty seconds. Press me to cancel." with heartbeat pulses	Yes

Trigger	Detection	Countdown wording	On by default
Hard fall	Firmware: free-fall then impact, gyroscope rotation check, 20 s stillness (spec §8.1)	"Did you fall? Press me or say I'm fine." then the same 30 s	Yes; sensitivity Normal / High (High keeps the gyroscope always on, about 11 mAh per day)
Voice	"Help me", "Bachao", "Kaapathu" and household-added words	Same as hold	Yes
Family raises it	Safety app: "Raise an alert for Amma" after a failed check-in	No countdown; the ladder starts and the pendant says "Priya is checking on you"	Yes
Wellness check	No press, voice or motion for 4 h during waking hours	"Are you alright? Press me." No press in 5 min sends a wellness push, not an SOS	Yes
Watch button	Wear OS / watchOS companion	Same as hold	V1.1

What an alert carries

Every push, SMS and automated call says who (the wearer's name), what (help requested, possible fall, voice), when (pendant time, and "happened 40 min ago" if late), where (map link, address, accuracy and age of the GPS fix), pendant battery, and how to respond: "Reply 1 = I'm on it, 2 = call me" by SMS, "press 1" on the call, one tap in the app. Messages are sent in each contact's chosen language.

Escalation ladder

1. T+0: tier 1 (up to 3 contacts) gets push, SMS and an automated call at once; the pendant says "Help is on the way. I've told Priya." (spec §7).
2. T+2 min without acknowledgement: tier 2 contacts are added; tier 1 is called again.
3. T+5 min: every contact including the Owner; calls repeat every 3 min for 20 min; push repeats every 10 min until resolved.
4. Any acknowledgement stops the calls, tells the wearer who is coming ("Priya is on her way"), and moves the incident to Acknowledged; the others are told who took it.
5. Resolution needs an outcome: real emergency, fall without injury, false alarm, or drill. Three false falls in 30 days prompt a sensitivity review.

Live incident room wireframe

```
| INCIDENT  .  Possible fall  .  Amma  .  started 14:52 (3 min ago)  .  Alerting tier 2                |
+-----------------------------------------+-----------------------------------------------------+
|  [ map: last fix 14:52, 12 m accuracy ]    |  Contacts                                                 |
|  12 Lakshmi Nagar 2nd St, Chennai          |   Priya (daughter)    push seen . SMS sent . call: ringing|
|  Gate code 4471 . spare key with Mrs. Rao  |   Ravi (son, US)      push sent . SMS sent . call: no ans |
|                                            |   Mrs. Rao (neighbour) SMS sent . call: answered, press 1 |
|  Pendant: battery 64%, connected           |                                                           |
|  Phone: at home, 1 min ago                 |  [ I'm on it ]   [ Open voice line to the pendant ]      |
|                                            |  [ Call Amma's phone ]   [ Call 112 ]   [ Share link ]   |
+-----------------------------------------+-----------------------------------------------------+
|  Timeline                                                                                          |
|  14:52:04  Fall candidate confirmed by pendant (stillness 20 s)                                      |
|  14:52:34  No response to "Did you fall?"; countdown ended; alerts sent to tier 1                    |
|  14:53:10  Priya opened the push                                                                    |
|  14:54:34  No acknowledgement in 2 min; tier 2 added; calling again                                 |
|  14:55:02  Mrs. Rao pressed 1 on the call: acknowledged                                             |
+-----------------------------------------------------------------------------------------------------+
|  Resolve:  ( ) Real emergency  ( ) Fall, no injury  ( ) False alarm  ( ) Drill   Note [          ] [Resolve] |
+-----------------------------------------------------------------------------------------------------+
```

"Open voice line" streams the pendant microphone and speaker through the Gateway to the browser so a family member can talk with the wearer; it needs the phone in range and is written to the audit log.

Safety app (no incident active)

The Safety app shows the ladder with each contact's last verification, the last drill date, fall sensitivity, voice-trigger words, wellness-check hours, the local emergency number derived from the home address, incident history with outcomes, and a "Run a drill" button that sends alerts marked DRILL.

Edge cases

- Phone out of range at trigger: the pendant sounds a loud alarm and says "I need help" every 30 s (spec §8.1); the event is stored and the tenant raises the incident on reconnect, marked late. Today's 2-hour offline rule is the safety net while the pendant is out of range.
- Phone has no internet: nothing reaches the tenant; the Gateway queues the event and, on Android, can still send one SMS from the phone itself to tier 1 if the household allowed it in setup.
- Tenant unavailable: the Gateway posts the alert to the control-plane relay, which carries only contact numbers and the location link, never household content.
- Wearer cancels a real fall: the family still receives a "fall detected, cancelled by Amma" note (spec §9 reports cancelled candidates).
- Accidental hold: a press in the countdown cancels; the pendant says "Cancelled"; nothing is sent.
- Contact's phone is on Do Not Disturb: Svaraa requests critical-alert entitlement on iOS and a bypass-DND channel on Android; SMS and calls are the fallback.
- SMS undeliverable: the call goes out immediately; a number that fails twice is flagged on the ladder.
- Two contacts acknowledge: both are shown; the first is the lead; the room shows who is calling whom.
- Pendant is on the charger, off the wrist: an SOS still alerts; the location is the phone's and the alert says the pendant was charging.
- Wearer abroad: the emergency number and message language follow the GPS country; the ladder stays the same.
- Battery at 5%: SOS and fall alerts still work (spec §6); the alert carries the battery level.
- Gateway killed by the OS: three drops in a day raise a Today item with the battery-optimisation fix; the household is offered the home-hub phone.
- Stale location: the alert shows the age of the fix and the home address as a fallback.
- Second incident while one is open: it is merged into the open incident with a new timeline entry.
- Location sharing outside incidents is off by default; a Wearer-managed household can turn on always-on location for the Owner.

---

12. Recite & Music

Svaraa's own devotional library plays on the pendant speaker; everything else (Spotify, YouTube Music, Apple Music) plays on the phone or a Bluetooth speaker with the pendant acting as a voice remote, exactly the split the hardware spec makes.

Two sources, two outputs

Source	Plays on	Works offline	Battery note
Svaraa devotional library (mantras, bhajans, aarti, chalisa, guided prayer, scripture readings, stories, sleep stories)	Pendant speaker (Opus 24 kbps over BLE), or the phone / a speaker if the wearer prefers	Yes: the Gateway caches the household's playlists on the phone	30 minutes on the pendant costs about 30 mAh, roughly half a typical day's budget; the app shows this beside the schedule
Third-party music	Phone speaker, Bluetooth speaker, Android LE Audio hearing aids; never the pendant	No	None on the pendant

Music integrations

Service	Connect	What voice control can do	Limits to state in the app
Spotify	OAuth in Connections	Search, play, pause, next, volume, choose the output device	Remote control needs Spotify Premium; the Spotify app must be installed on the Gateway phone
YouTube Music	Google OAuth in Connections	Search and open a result on the phone; pause, next and volume through the phone's media controls (Android; limited on iOS)	No official playback API; the app says "I've opened YouTube Music with old Tamil songs" rather than promising playback
Apple Music	MusicKit on an iOS Gateway phone	Search, play, pause, next, volume	iOS Gateway only

All three are verified at build time; the Connections app shows a plain-language capability line per service so nobody is surprised by a limit.

Recite app wireframe

```
| Recite  .  Library for Amma                    Language: [Tamil v] [Hindi] [Sanskrit]   Tradition: [Hindu v] |
+-------------------------------------------------------------------------------------------------------------+
|  Now playing on pendant:  Hanuman Chalisa  .  4:12 / 9:40   [pause] [next] [vol -] [vol +]   Output: pendant  |
+-------------------------------------------------------------------------------------------------------------+
|  Amma's playlists                          |  Browse                                                        |
|   Morning (Suprabhatam, 3 items, 24 min)   |   Aarti (12)   Chalisa (6)   Bhajans (140)   Guided prayer (9)      |
|   Evening aarti (2 items, 8 min)           |   Stories (48)   Sleep stories (12)   Scripture readings (30)       |
|   Sleep (Vishnu Sahasranama, 22 min)       |                                                                |
|   [+ New playlist]                         |   Hanuman Chalisa       9:40   Tamil     [Play on pendant] [Add]   |
|                                            |   Om Jai Jagdish Hare   6:05   Hindi     [Play on pendant] [Add]   |
|  Scheduled                                 |   Lalitha Sahasranama  28:10   Sanskrit  [Play on pendant] [Add]   |
|   06:00 Morning (with prayer reminder)     |                                                                |
|   18:24 Evening aarti (sunset)             |  Family sends                                                  |
|   21:30 Sleep, volume 40%                  |   Anjali sent "Amma's favourite bhajan"  [Play now] [Schedule]     |
+--------------------------------------------+----------------------------------------------------------------+
```

Music app wireframe

```
| Music  .  Spotify (Premium, connected)  .  YouTube Music (connected)  .  Apple Music (not connected)        |
+-------------------------------------------------------------------------------------------------------------+
|  Playing on Amma's phone via Spotify:  "Kanne Kalaimane"  .  Ilaiyaraaja   [pause] [next] [vol]             |
|  Output: [Phone speaker v]  options: Phone speaker . Living-room speaker (Bluetooth) . Hearing aids          |
+-------------------------------------------------------------------------------------------------------------+
|  Quick voice phrases Amma can say                  |  Family picks                                          |
|   "Play old Tamil songs"        -> Spotify playlist |   Ravi: "Sunday morning mix"   [Play now] [Send to Amma]|
|   "Play MS Subbulakshmi"        -> Spotify artist   |                                                        |
|   "Play bhajans on YouTube"     -> opens YT Music   |  Rules                                                 |
|   "Stop" / "Louder" / "Next"    -> phone controls   |   Explicit content: off . Night volume cap: 50% after 10 pm|
+-------------------------------------------------------------------------------------------------------------+
```

Rules

- Playback pauses for a reminder and resumes after it; a voice question pauses playback for the reply; an SOS stops playback outright.
- A recitation attached to a prayer reminder plays on the pendant if it is worn and connected, otherwise on the phone, and says where it is playing.
- Windows in which the pendant was playing are excluded from transcription in continuous mode, so the library is never transcribed as conversation.
- Family members can send a library item or a music link to play now (with a spoken "Anjali sent you a bhajan") or schedule it; the wearer can say "not now".
- Volume is capped at the night level during quiet hours; the pendant's firmware limiter protects the speaker (spec §4).

Edge cases

- Spotify account is not Premium: the Connections card explains it, and voice requests answer "Spotify needs Premium to play from here; shall I open it on your phone?"
- Music token expired: Today shows a reconnect card; voice requests say the service needs the family to reconnect it.
- No music app on the Gateway phone: setup blocks the connection with an install link.
- Song not in the library: the assistant offers the connected music service or says it is not available; nothing is played from an unknown source.
- Content not available in the region: the service's error is translated into one sentence.
- Pendant battery below 15%: the pendant asks to be charged (firmware); recitations longer than 10 minutes are routed to the phone. At 5% playback stops (spec §6).
- Pendant on the charger at prayer time: the recitation plays on the phone and the reminder still fires on the pendant.
- Loud room: pendant speaker peaks near 83 dB at 10 cm; the assistant offers to move playback to the phone or speaker.
- Library content licence lapses: the item disappears from browse, scheduled playlists skip it and the family is told once.
- Two requests in a row: the newer request replaces the current playback; "queue this" adds instead.
- Wearer asks for something explicit or unsuitable: the household's explicit filter applies to third-party services where they support it.

---

13. News, briefings & engagement

The pendant gives the wearer a 90-second morning briefing in their own language, and the family a set of small ways to be present during the day: voice notes, check-ins, prompts and shared life stories.

Morning briefing

Part	Source	Length
Greeting, date, festival or observance today	Prayer engine calendar	1 sentence
Weather for the home location	Weather provider	1 sentence
Today's appointments and important reminders	Reminder engine	Up to 3 items
Family news	Voice notes received, visits planned, birthdays this week	Up to 2 items
Headlines	3 to 5 items from the household's chosen sources, summarised in the wearer's language, each ending with the source name	1 sentence each
Line of the day	Scripture verse or saying from the wearer's tradition	1 sentence
Close	"Want more on any of these?" opens the microphone for 6 s	

The briefing plays on the pendant at the set time if it is worn and connected, otherwise on the phone; it waits up to 2 hours for the first press or movement if the wearer is still asleep, then skips. "What's the news?" plays the headline part at any time.

News rules

- Sources are a curated list per language and region (national dailies, regional-language dailies, public broadcasters); the family can add or remove sources but not add social media.
- Every headline is summarised only from the fetched article, never from model memory, and carries its source and publication time; "more on that" reads a longer summary of the same article.
- Calm mode (default on) softens or omits detail of crime, accidents and disasters and never reads casualty counts; the family can turn it off.
- Topics: local, national, community and spiritual, sport, weather; politics is summarised neutrally with no opinion, and the assistant declines to argue a political position.
- A distressing follow-up question gets a short factual answer and an offer to change the subject.

Engagement features

Feature	Family side	Wearer side	Limits
Voice note	Record in the web app or Gateway, up to 2 minutes	Pendant says "Anjali sent a voice note", plays it, opens the microphone for a reply	10 notes per member per day; queued if the pendant is offline and played with "from earlier"
Check-in	Scheduled in Routines or sent now: "How are you feeling?"	Spoken answer transcribed and shown on Today	An unanswered check-in twice in a day raises an attention item
Ask a question	A grandchild types or records a question	Wearer answers by voice; the answer goes back as audio and text	Same daily limit as notes
Life stories	Family adds prompts ("Tell me about your wedding day") or picks from a library of 200	Asked at a chosen time; the answer is recorded and kept as a story the wearer can share with the family	Wearer chooses sharing per story; stories are excluded from the transcription cap

Feature	Family side	Wearer side	Limits
Line of the day, quizzes and riddles	Settings only	Short voice games from the wearer's tradition and language	Off by default in Family-managed households until the wearer asks
Family calls	Schedule a weekly call as a reminder	"Ravi's call in 10 minutes"	The call itself happens on the phone

News app wireframe

```
| News  .  Amma's briefing                              Language: [Tamil v]   Time: [07:05]   Calm mode: [on] |
+-------------------------------------------------------------------------------------------------------------+
|  Today's briefing (played 07:05 on the pendant, 1 min 32 s)          [Play on pendant again] [Read aloud]  |
|   Good morning, Amma. Today is Saturday, Ashtami.                                                           |
|   Chennai: 31 degrees, light rain by evening.                                                               |
|   Today: blood-pressure tablet at nine; the cab for Dr. Menon comes at 9:15 tomorrow.                       |
|   Anjali sent you a voice note last night.                                                                  |
|   1. Metro line to Poonamallee opens next month (The Hindu, 06:10)                                          |
|   2. Temple festival at Mylapore begins Sunday (Dinamalar, 05:40)                                           |
|   3. Light rain expected across Tamil Nadu this weekend (Dinamalar, 06:00)                                  |
|   Line of the day: ...                                                                                      |
+---------------------------------+---------------------------------------------------------------------------+
|  Topics                         |  Sources                                                                  |
|   [x] Local   [x] National      |   [x] The Hindu   [x] Dinamalar   [ ] Dina Thanthi   [x] Doordarshan      |
|   [x] Community and spiritual   |   [+ Add a source]                                                        |
|   [ ] Sport   [x] Weather       |                                                                           |
+---------------------------------+---------------------------------------------------------------------------+
```

Routines app wireframe

```
| Routines  .  Amma                                                                                           |
+-------------------------------------------------------------------------------------------------------------+
|  07:05  Morning briefing           daily       on pendant, else phone                  [Edit]       |
|  14:00  Check-in "How are you feeling?"   daily                                         [Edit]       |
|  17:00  Life story prompt          Tue, Thu    next: "Your first day at the bank"       [Edit]       |
|  18:00  Evening digest to family   daily       Priya, Ravi                             [Edit]       |
|  Quiet hours  22:00 to 06:00       Safety alerts always; important medication may speak   [Edit]       |
|  Wellness check  no activity for 4 h between 07:00 and 21:00                           [Edit]       |
+-------------------------------------------------------------------------------------------------------------+
```

Edge cases

- No internet at briefing time: the briefing plays without news and weather from the Gateway's cached reminders and says the news will come later.
- A source is down or returns only stale items: it is skipped; two days of silence from a source raise a Connections notice.
- Duplicate stories across sources: de-duplicated by headline similarity; the first source is credited.
- Article in a different language than the wearer's: summarised in the wearer's language with the source named.
- Wearer asks about news the briefing did not carry: the assistant searches the curated sources only and says when it has nothing.
- Voice note arrives at night: held until quiet hours end unless the sender marks it urgent, which is limited to Owner and Caregivers.
- Too many notes: beyond the daily limit the sender is told; notes never interrupt a reminder or an incident.
- A life-story prompt lands on a hard day: the wearer can say "not today" and the prompt moves a week; the family is not told the reason.
- Life story shared then withdrawn: the wearer can unshare; copies already played are not recalled, and the app says so.
- Quiz answers are never scored in the digest; games leave no record beyond "played".

---

14. Connections: integrations (Riddle)

Every integration runs in the phone app or the tenant, never on the pendant, so adding a service never touches firmware; the Connections app shows each one with a plain state and what it is used for.

Integration catalogue

Integration	Runs in	Used for	Auth	V1
Google Calendar, Microsoft 365, Apple iCloud (CalDAV)	Tenant	Appointments become reminders; events tagged "record" start scheduled recording; visit dates feed the briefing	OAuth or app password	Yes
Apple Health (iOS Gateway) / Health Connect (Android Gateway)	Gateway	Reads steps, heart rate and sleep from the wearer's phone or watch; writes reminders taken and falls (spec §8.3)	OS permission per data type	Yes, only when the Gateway phone is the wearer's own
SMS and automated calls	Tenant via a telephony provider (Twilio, Exotel or MSG91 class; India DLT-registered templates)	SOS alerts, reminder escalations, contact verification	Svaraa-managed	Yes
Push notifications	Gateway (APNs, FCM)	Everything routed to family phones; critical-alert channel for Safety	OS permission	Yes
Spotify, YouTube Music, Apple Music	Gateway plus tenant	Section 12	OAuth	Yes

Integration	Runs in	Used for	Auth	V1
Phone contacts	Gateway	"Call Ravi" and family names in summaries	OS permission	Yes
Home hub (spare phone or tablet running the Gateway in hub mode)	Gateway	Always-on bridge at home; primary when in range	Household link code	Yes
Hearing aids	Phone OS (Android LE Audio; iOS hearing-device support)	Assistant replies and library audio routed to hearing aids	OS pairing	Partial (spec §8.3)
WhatsApp alerts	Tenant via WhatsApp Business API	Optional alert channel for families who do not read SMS	Recipient opt-in	V1.1
Apple Watch / Wear OS companion	Gateway	Reminders mirrored, SOS button, status complication	Watch pairing	V1.1
Apple Find My / Google Find Hub	Pendant firmware (MFi enrolment)	Lost pendant beyond family phones	Programme enrolment	V2; V1 uses BLE proximity from family phones (section 15)

Connections app wireframe

```
| Connections  .  Amma's household                                                                                  |
+-------------------------------------------------------------------------------------------------------------------+
|  Calendar          Google Calendar (priya@...)      Connected . last sync 5 min ago     [Settings] [Disconnect]|
|                    Uses: appointments, visits, record tags . Calendars: Amma, Family (2 of 5)                    |
|  Health            Health Connect on Amma's phone   Connected . steps, sleep            [Permissions]      |
|                    Heart rate: not granted . Writes: reminders taken, falls                                       |
|  Music             Spotify (Premium)                Connected                           [Disconnect]       |
|                    YouTube Music                    Needs attention: sign in again      [Reconnect]        |
|                    Apple Music                      Not connected (needs an iPhone Gateway)                    |
|  Alerts            SMS and calls                    Healthy . 2 of 2 numbers verified   [Test]             |
|                    Push                             Priya: on . Ravi: on . Anjali: off (phone setting)     |
|  Home hub          Old tablet in the living room    Online . plugged in . primary at home [Settings]       |
|  Hearing aids      Not set up                                                            [How to]           |
|  WhatsApp          Coming in V1.1                                                                             |
+-------------------------------------------------------------------------------------------------------------------+
```

Health app

The Health app shows what connected sources report (steps, sleep and heart rate from the wearer's phone or watch) beside what Svaraa itself records (reminders taken, fall events, check-in answers); it draws no medical conclusions and shows a source label on every number. An activity estimate from the pendant's motion sensor and barometer (floor changes, spec §4) is V1.1, pending the freedom-to-operate review the hardware spec calls for. The wearer can opt in to hearing yesterday's steps in the morning briefing.

Home hub rules

- The pendant bonds with up to three Gateway phones; it connects to one at a time and prefers the hub when both are in range, so a family member's phone leaving the house never takes the link with it.
- Hub mode keeps the screen off, ignores Health and Contacts, and is expected to stay plugged in; a hub that goes offline for 10 minutes raises a Today item.
- Location during an incident comes from the hub's stored home address when the hub is the connected Gateway.

Calendar rules

- Only calendars the family ticks are read; events are stored in the tenant as appointment records with the source event id; private events are read only if the calendar owner allows it.
- An event becomes a reminder pair (the evening before at 8 pm, and 1 hour before); the family can change the lead times per calendar.
- Times are converted to the wearer's zone; an all-day event becomes a morning-briefing line, not a reminder.

Edge cases

- OAuth token expires or is revoked: the card turns to Needs attention with a one-tap reconnect; dependent reminders keep their last known times and are marked "from calendar, not refreshed".
- The Gateway runs on a daughter's phone: Health is disabled with an explanation, so her steps are never recorded as the wearer's.
- Health Connect not installed (older Android): the card links to the install and works without it.
- SMS templates in India need DLT approval: all alert and verification texts are fixed templates with variable slots; free-text is never sent by SMS.
- International family numbers: alerts go out in E.164 format; a number the provider cannot reach is flagged on the ladder within a minute of the first attempt.
- Two calendars carry the same appointment: de-duplicated by time and title; the family sees one reminder pair.
- Hub and phone both in range and both connected to the internet: the pendant stays with the hub; the phone shows "connected via hub".
- Hearing-aid latency makes the assistant's reply lag: the pendant's tick still confirms the press; the reply is not repeated on the pendant speaker.
- A connected service changes its API or terms: the adapter is versioned in the tenant and the card shows "limited" rather than failing silently.

15. Pendant management & device health

The pendant has no screen and no lights, so the Pendant app is where the family sees everything the wearer only hears: battery, charging, storage, mute, connection and firmware, with one action per problem.

Pendant app wireframe

```
| Pendant  .  "Amma's pendant"  SL1-7F2A                                                  [Find my pendant] |
+-------------------------------------------------------------------------------------------------------+
|  Battery 70%  (about 3 days left)     Charging: no . last charged Wed 22:10 (3 days ago)  [Send charge nudge]|
|  Connected via: Living-room hub . seen 1 min ago       Microphones: on       Recording: off                 |
|  Storage: 2 recordings waiting (18 MB) . 3% of memory used                                                  |
|  Firmware 1.0.3 (latest)  . next check tonight                                                              |
|  Battery, last 7 days:  a line from 100% on Wed to 70% today, with charge points marked                     |
+-----------------------------------------------+-------------------------------------------------------+
|  Gestures                                     |  Fall detection                                       |
|   Single press: talk / acknowledge   (always) |   Sensitivity  (o) Normal  ( ) High (+11 mAh a day)   |
|   Hold 3 s: help                     (always) |   Wellness check after [4] h without activity         |
|   Double press: spoken status        [on]     |   Last drill: 12 Sep . [Run a drill]                  |
|   Triple press: microphones off      [on]     |                                                       |
|  Sound and touch                              |  Recording                                            |
|   Volume  [-] o------o------ [+]  (chime)     |   Mode: On request . change in Settings > Privacy     |
|   Haptic strength  ( ) Soft (o) Normal        |   10-minute chime while recording: on (required)      |
|   Spoken language for chimes: Tamil           |                                                       |
+-----------------------------------------------+-------------------------------------------------------+
|  Care                                                                                                   |
|   [Restart guide]  [Dry-the-port tip]  [Replace pendant]  [Mark as lost]  [Factory reset (Owner)]       |
+-------------------------------------------------------------------------------------------------------+
```

Battery and charging rules (spec §6)

Level or event	Pendant does	App does
20%	Nothing new	Warns the family on Today and by push once
15%	Says "Please charge me soon" once a day	Attention item stays until charged
5%	Stops recording and playback; keeps button, fall alerts and reminders	Red state; voice questions disabled with an explanation
Plugged in	Says "Charging, 40 percent"	Shows charging and clears the charge items
Not charged for 3 days	Nothing	Attention item and push to Caregivers
Not connected for 2 h in waking hours	Nothing	Attention item and push; suggests the hub if this repeats
Above 40 °C while charging	Charges slower (JEITA)	"Charging slowly because it is warm"

The first charge of each day in the first two weeks is preceded by a spoken "Dry the port first", then once a week; the app shows the same tip, because sweat in the USB-C port is a listed hardware risk.

Firmware updates

- Updates are signed and installed by MCUboot with rollback; the pendant must be charging with battery at 50% or more (spec §9).
- Rollouts are staged (1%, 10%, 100% of households) with an "early updates" opt-in; release notes are shown in plain words.
- The default is "update tonight while charging"; a failed update retries the next night and the app says which version is running.

Find my pendant

- Every family phone running the Gateway reports when it sees the pendant's BLE advertising, so the app can say "near Priya's phone, 3 minutes ago".
- "Make a sound" plays a chime for 30 s when connected; the last-connected location is the Gateway phone's position at that moment.
- Apple Find My and Google Find Hub are V2 (section 14).

Lost, replaced, reset

- Mark as lost: voice questions and recording are disabled; if the pendant reconnects, the family is alerted with the location; reminders and SOS keep working in case the wearer has it after all.
- Replace pendant: pairing a new pendant copies the wearer profile, schedule and settings; the old pendant is unbonded and wiped on its next connection.
- Factory reset (Owner only, typed confirmation): unbonds and wipes the pendant; tenant data is untouched.

Edge cases

- Storage nearly full because the phone was away: the app shows the queue and an estimate; the pendant refuses new recordings at 100% with a double falling chime.
- Clock reset after a flat battery: the Gateway sets the clock on connect; recordings made before that are flagged "approximate time" (section 8).
- Pendant unresponsive: the restart guide (hold 10 s while plugged in) is shown with a picture; the watchdog also restarts it on its own.
- Battery capacity below 80% of new (from cycle count and fuel gauge): the app shows "battery ageing, about 2 days per charge" and offers a replacement.
- Charging-port fault detected by the PMIC (over-voltage or no current on a known cable): "Check the charging port for moisture or dirt".
- New pendant in ship mode: it wakes only when plugged in; the pairing step says so.
- Double or triple press disabled by the family: the app shows that spoken status and mute are off and how to re-enable.
- Two pendants: the app shows one page per pendant with the wearer's name; settings are never shared between them.
- Pendant worn by someone else (a visitor tries it): the assistant answers as usual; a session with no wearer voice is not summarised.

---

16. Family circle & notifications

The family circle is the list of people who can see or be told about the wearer, and notification routing decides who hears what, on which channel, at what hour, so a son in San Jose gets the doctor summary at breakfast and the fall alert at 3 am.

Family app wireframe

```
| Family  .  Amma's circle                                                                                  [+ Invite] [+ Emergency contact] |
+-----------------------------------------------------------------------------------------------------------------------------------------+
|  Member            Role        SOS tier   Alerts                          Last active         Actions                      |
|  Priya (daughter)  Owner       Tier 1     push, SMS, call . all           10 min ago          [Edit] [Voice note]          |
|  Ravi (son)        Caregiver   Tier 1     push, SMS, call . US hours      yesterday           [Edit] [Voice note]          |
|  Anjali (17)       Viewer      -          push . family only              2 h ago             [Edit] [Voice note]          |
|  Amma              Wearer      -          pendant                         12 min ago          [Simple Mode settings]       |
|  Mrs. Rao          Emergency   Tier 2     SMS, call . verified 14 Aug     -                   [Re-verify] [Remove]         |
|  Dr. Menon         Contact     -          none (address book only)        -                   [Edit]                       |
+-----------------------------------------------------------------------------------------------------------------------------------------+
|  Pending: invite to lakshmi@... (Caregiver) sent 2 days ago  [Resend] [Revoke]                                                           |
|  Amma can hear: "Priya checked on you today" (presence line in Simple Mode)  [on]                                                         |
+-----------------------------------------------------------------------------------------------------------------------------------------+
```

Invitations

1. Owner enters an email or phone and picks a role (and SOS tier if Caregiver); the invite carries the household name and expires in 7 days.
2. The invitee opens the link, verifies by OTP, and creates or signs in to an account; the role is applied on join.
3. Emergency contacts get a verification SMS with a 4-digit code the Owner types in; a voice call reads the code if SMS fails. They are told what they will receive and how to opt out by replying STOP.
4. Contacts are re-verified every 6 months by a test message; a failure flags the ladder on Today.

Notification routing

Event	Owner and Caregivers	Viewer	Emergency contact	Wearer	Quiet hours
SOS, fall, wellness push	Critical push, SMS, automated call per tier	Push if opted in	SMS and call per tier	Pendant speech	Never suppressed
Important reminder unacknowledged	Push with actions	No	No	Pendant repeats	Never suppressed
Device (battery 20%, not charged 3 days, offline 2 h)	Push, grouped once per issue	No	No	Pendant speech (battery only)	Held until morning except offline 2 h
New session summary	Push if Recall access, or in the digest	No	No	On request by voice	Held
Evening digest	Push at digest time (member's own zone optional)	Status-only digest	No	On request	Held
Voice note, check-in answer, question answered	Push to the sender	Push to the sender	No	Pendant plays	Held unless marked urgent by Owner or Caregiver
Account, billing, provider issues	Email and push to Owner	No	No	No	Held

- Each member sets channels, quiet hours in their own time zone, and "only safety" mode; safety events ignore all of these.
- Push tokens that fail twice fall back to SMS for safety events and email for everything else, and the member is asked to reopen the app.
- Alerts are grouped: one push per issue, updated in place, never a stream of repeats, except the safety ladder which is designed to repeat.

Presence for the wearer

When a family member opens Today or plays a session, the wearer can hear "Priya checked on you today" in the morning briefing and see it in Simple Mode. This is on by default in Wearer-managed households and off in Family-managed ones; the family sees which setting is active.

Edge cases

- Member in another time zone: digest and non-safety pushes follow the member's zone; the SOS ladder ignores zones.
- Wrong person invited: revoke before or after join; a revoked member's sessions are ended within a minute.
- Role downgraded: Recall access ends at once; open session tabs close with a notice.
- Last Owner wants to leave: must transfer ownership first; the transfer needs the new Owner's acceptance.
- Too many Caregivers on tier 1: the app caps tier 1 at 3 and suggests tiers to avoid everyone being called at once.
- SMS reply from an unknown number: ignored and logged; only verified numbers can acknowledge by keyword.
- A Viewer taps "I'm on it": not offered; a Viewer sees the incident status only.
- Member's phone lost: the account page lists devices with a sign-out-everywhere action.
- Duplicate contact numbers across members and emergency contacts: merged with a notice so one person is not called twice.
- A member is the Wearer in another household: their own pendant's alerts never mix with this household's; the switcher shows both.

---

17. Settings, privacy, consent & data controls

Privacy in Svaraa is a set of recorded decisions the wearer can hear and revoke by voice, not a policy page: every consent has a record, every look at a transcript is logged, and the household can take all its data out or destroy it.

Settings map

Group	Contains	Who
Household	Name, wearers, wearer time zone, data region (fixed after provisioning; a move is export plus re-provision)	Owner
Privacy and consent	Privacy mode, recording mode, consent records, transcript sharing, retention, location sharing, presence line, third-party recording notice	Owner, Wearer
AI	Brain, speech engines, keys, fallback toggle, usage meter and caps, disclosure history	Owner
Notifications	Own channels, quiet hours, only-safety mode	Each member
Billing	Plan, allowance, invoices, payment method	Owner
Data	Export, delete sessions, delete household, audit log, support access token	Owner, Wearer (own data)
Account	Email and phone, password, two-factor, devices and sessions, sign out everywhere	Each member
Accessibility	Simple Mode default, text size, contrast, reduced motion, spoken confirmations	Each member

Privacy and consent wireframe

```
| Settings  .  Privacy and consent                                                    Privacy mode: Wearer-managed |
+-------------------------------------------------------------------------------------------------------------------+
|  Recording                                                                                                        |
|   Mode: (o) On request   ( ) Continuous during waking hours     10-minute chime: always on                       |
|   Consent: Amma said "I agree" on 14 Aug 2026, 10:12 (voice record, v2 of the notice)   [Play] [Ask again]       |
|  Sharing                                                                                                          |
|   Family can see: (o) Summaries and action items   ( ) Full transcripts and audio   ( ) Nothing                  |
|   Consent: Amma, 14 Aug 2026 (voice)        Amma can change this any time by saying "Svaraa, privacy"             |
|  Retention                                                                                                        |
|   Audio: [30 days v]   Transcripts and summaries: [until deleted v]   Voice-question answers: 30 days             |
|  Location                                                                                                         |
|   Shared during incidents: always   Shared at other times: [off v]                                               |
|  Notices                                                                                                          |
|   Recording notice for visitors: [Print door card]  [Spoken notice when a new voice is heard: off]                |
|   Where your data is processed: tenant in Mumbai; speech by Sarvam (India); text by Claude (US)   [Details]       |
+-------------------------------------------------------------------------------------------------------------------+
|  Audit log (last 7 days)                                                                                          |
|   26 Sep 14:40  Ravi read "Phone call with Ravi" summary                                                          |
|   25 Sep 21:02  Priya played audio of "Visit from Mrs. Rao"                                                       |
|   25 Sep 09:15  Schedule pushed to pendant (v41)                                                                  |
+-------------------------------------------------------------------------------------------------------------------+
```

Consent model

Decision	Who consents	How it is captured	Re-asked when
Recording on request	Wearer (or Owner attestation in Family-managed mode)	Voice "I agree" on the pendant, typed acceptance, or attestation; notice version stored	Notice text changes
Continuous recording	Wearer only, or attestation	Voice record; jurisdiction note shown (two-party consent regions)	Every 6 months, and on any provider change
Transcript sharing level	Wearer	Voice or Simple Mode	Wearer changes it; Owner cannot raise it
AI provider and processing regions	Owner, with the wearer told by voice	Typed acceptance of the disclosure	Provider or routing changes
Location outside incidents	Wearer	Voice or Simple Mode	Never automatically

The wearer's voice always wins: "Svaraa, stop sharing my recordings" or "Svaraa, stop recording" applies after one spoken confirmation, and the Owner is told what changed, not asked.

Retention defaults

Data	Default	Notes
Raw audio	30 days	7, 90 or keep; pinned sessions keep audio
Transcripts, summaries, action items	Until deleted	Wearer can delete own sessions; Owner can delete any
Voice-question answers	30 days	Questions stay in the voice log
Incidents and drills	2 years	Outcome and timeline only; audio from the voice line is not stored
Audit log	2 years	Read-only
Consent records	Life of the household plus 1 year	Kept in minimal form (who, what, when, version) in the control plane after deletion

Export, delete and offboarding

- Export: a full archive (transcripts and summaries as PDF and JSON, audio files, reminders, contacts, audit log) as an encrypted zip with a password shown once; ready within 24 hours, link valid 7 days. Any single session exports as a PDF.
- Delete a session: audio, transcript, summary and search entries go at once; action items created from it keep their text and lose the link.
- Delete the household: 30-day grace with one-tap restore, then the data keys are destroyed, which makes every stored file unreadable; the pendant is wiped on its next connection.
- Closing a household after a death: a separate, gently worded flow. It stops every reminder, nudge and alert immediately so nobody receives "please charge me", keeps sessions and life stories read-only for the family for a chosen period (30 days to 1 year), offers the export, and then deletes. Billing stops on the day the flow starts.

- Support access: the Owner generates a token that gives Svaraa support read-only access for 24 hours; every use is in the audit log.

Compliance touchpoints (legal review required)

- India's Digital Personal Data Protection Act: notice in the wearer's language, purpose-limited consent records, data principal rights (access, correction, erasure) served by the export and delete flows, and a grievance contact in Settings.
- GDPR for EU-region households: the same rights plus data-processing disclosure per provider.
- Not a medical device: no diagnosis, no dose advice, and marketing copy per the hardware spec's rule.
- Minors as Viewers: no transcripts, no location, minimal profile.

Edge cases

- Owner switches provider: the disclosure is re-accepted and the wearer hears a one-line notice; continuous mode pauses until the wearer re-agrees if the speech provider changed.
- Wearer withdraws consent in a Family-managed household: applied at once; the Owner sees a note and can discuss it with the wearer, not override it.
- No consent yet and the wearer cannot give it: recording stays off; every other feature works.
- Export requested during an incident: queued until the incident resolves.
- Delete with recordings still on the pendant: the wipe on next connection removes them; the app shows "2 recordings still on the pendant" until then.
- Account compromise suspected: sign out everywhere rotates every session and Gateway token; the pendant bond is unaffected.
- Region move requested: export, delete, re-provision in the new region, re-pair the pendant; the app explains the four steps before starting.
- A session is deleted while a Caregiver has it open: the tab closes with a notice; no cached copy remains in the browser.

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18. Degraded-mode matrix & global edge cases

The system is a chain (pendant, phone, tenant, providers) and each link can fail on its own; the rule is that reminders, the fall countdown and the SOS countdown never leave the pendant, so the worst failure delays an alert rather than losing it.

What still works when a link fails

Failure	Voice questions	Reminders	SOS and falls	Recording	Family alerts	What the wearer hears	What the family sees
Pendant out of BLE range (10 to 20 m)	No	Yes, on the pendant	Countdown runs; local loud alarm; event stored and sent on reconnect	Stored on pendant, up to 17 h	Late, marked with the delay	"I can't reach your phone"	Amber after 30 min, attention item at 2 h

Failure	Voice questions	Reminders	SOS and falls	Recording	Family alerts	What the wearer hears	What the family sees
Pendant flat (0%)	No	No	No	No	None from the pendant	Nothing	"Pendant off since 14:10"; the 2 h and 3-day rules escalate
Gateway app killed or phone off	No	Yes	As out of range	As out of range	Late	"I can't reach your phone"	Same as out of range; a fix card if the OS killed the app
Phone has no internet	Tier-0 only from the Gateway cache	Yes	Countdown and alarm; Android can send one SMS directly if enabled	Stored on the phone, uploaded later	Delayed; Android SMS fallback	"I can't think right now; your reminders still work"	"Phone offline" banner; last known state
Tenant degraded (jobs or storage)	Yes if API up	Yes	Yes; alerts via tenant	Uploads queue; transcription delayed	Yes	Normal	"Summaries delayed" banner
Tenant down	Tier-0 only	Yes (cached schedule for 7 days)	Countdown and alarm; Gateway posts to the control-plane relay with numbers and location only	Stored on phone	Via relay, without household content	"I can't think right now"	Static outage page with the emergency numbers
AI provider down	Tier-0; fallback provider if allowed	Yes	Yes	Recording continues; transcription queued	Yes	Section 4 wording	Provider banner in the companion panel
Telephony provider down	Yes	Yes	Push still goes; SMS and calls retried on a second provider	Yes	Push only until recovered	Normal	Safety banner: "calls delayed, push working"
Music or calendar service down	Yes	Yes (last synced)	Yes	Yes	Yes	"Spotify isn't answering right now"	Needs-attention card
Control plane down	Yes	Yes	Yes (tenant routes alerts)	Yes	Yes	Normal	Sign-in and billing unavailable; existing sessions continue

Global edge cases

- Clocks: the pendant clock is set on every connect; recordings before a set are flagged; all schedules are stored in the wearer's zone; daylight-saving changes in a family member's zone shift only that member's notifications.
- Travel: when the wearer travels with a family member, that member's phone becomes the Gateway for the trip ("Travel mode", one tap); prayer times and emergency numbers follow the GPS country; the hub at home reports "pendant away".
- Hospital stay: "Pause mode" with a reason and an end date suppresses fall detection and wellness checks (hospital beds and lifts cause false candidates), keeps SOS and reminders, and tells the family who paused it.
- Not worn: no motion and no press all day suggests the pendant is on a table; Today shows "not worn today?" without treating it as a safety event.
- Power cut at home: the hub goes offline; the wearer's own phone, if present, takes over; the family sees "hub offline" once, not every hour.
- Two caregivers edit the same reminder: last write wins with a notice to the other; a schedule push is never sent half-edited.
- Duplicate events after retries: every pendant event and file carries an id; the tenant is idempotent on them.
- Two pendants, one phone: each pendant has its own bond, profile and schedule; the Gateway multiplexes both.
- Languages: UI in English, Hindi and Tamil at V1 launch with the wearer's spoken language independent of the UI language; Indic numerals follow the UI locale; right-to-left is not needed for the V1 languages.
- Low-end phones: the Gateway must run on a 3 GB RAM Android phone; the web app must work in Safari and Chrome from the last two years and in the Gateway webview.
- Very long absence of the family from the app: nothing changes for the wearer; the weekly digest email carries the essentials.
- Wearer with hearing loss or a soft voice: haptic-first reminders, phone speech, and a microphone gain setting per wearer.
- Wearer who forgets the gestures: the printed card, a spoken reminder of the two gestures on the first press each day for the first two weeks, and a family voice note template "how to use your pendant".

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19. Non-functional requirements, security, analytics & metrics

The alert path is the number that matters: an SOS must leave the tenant within 3 seconds of arriving and the path must be exercised every night by a synthetic drill; everything else is sized around a 10,000-household first run.

Performance and availability

Requirement	Target
Voice question, press to first spoken word	2.0 s p50, 3.5 s p95 (section 4)
SOS alert dispatch after the tenant receives the event	Under 3 s to first push, SMS and call request
Schedule push after the pendant connects	Under 10 s

Requirement	Target
Today live update after an event reaches the tenant	Under 5 s
Web app first load on 4G	Under 2 s to Today; under 4 s in the Gateway webview on a 3 GB phone
Transcription of a 10-minute on-request file	Summary ready within 3 min of upload
Continuous-mode backlog	A full day processed within 1 h of upload
Tenant availability	99.9% monthly; alert path 99.95% including the control-plane relay
Backups	Encrypted in-region; recovery point 15 min, recovery time 1 h
Nightly synthetic drill	A test incident per region through push, SMS and call providers to Svaraa test numbers; failure pages the on-call

Security

- Sign-in with OTP or password plus optional two-factor; two-factor is required for Owners of Family-managed households.
- Tenant isolation: separate database, object-store prefix, encryption keys and network policy per household; no shared query path across tenants.
- Transport TLS 1.3; BLE LE Secure Connections with bonding; signed firmware; per-file audio encryption before upload; field-level encryption for transcripts and contacts.
- Content is never in logs; logs carry ids and timings only; analytics never carry transcript text or audio.
- Transcript text is data, never instructions: nothing said in a recording can trigger an action; actions come only from a press-to-talk turn by the wearer or a confirmed companion-panel card. This closes the "a visitor tells Svaraa to share everything" injection path.
- Rate limits on voice turns, notes and invitations per member; abuse of the SOS ladder by a member is visible in the audit log and revocable by the Owner.
- Penetration test and a privacy review before launch; dependency and provider adapters versioned and pinned.

Analytics events (no content, per-household opt-out)

Event	Properties	Purpose
onboarding_step_completed	step, duration, path (family or wearer)	Find where setup stalls
pendant_connected, pendant_disconnected	gateway kind (phone or hub), gap length	Range and reliability
voice_turn	intent class, tier, provider, latency, success	Assistant quality and cost
reminder_fired, reminder_acked, reminder_missed	type, delay to acknowledgement, channel	Adherence and escalation tuning
session_created	kind, duration, mode, language	Recall usage and cap planning
summary_viewed, summary_edited	role, edit kind	Summary quality
incident_started, incident_acked, incident_resolved	trigger, tier reached, time to acknowledgement, outcome	Safety performance and false-alarm rate
drill_completed	time to acknowledgement	Ladder health

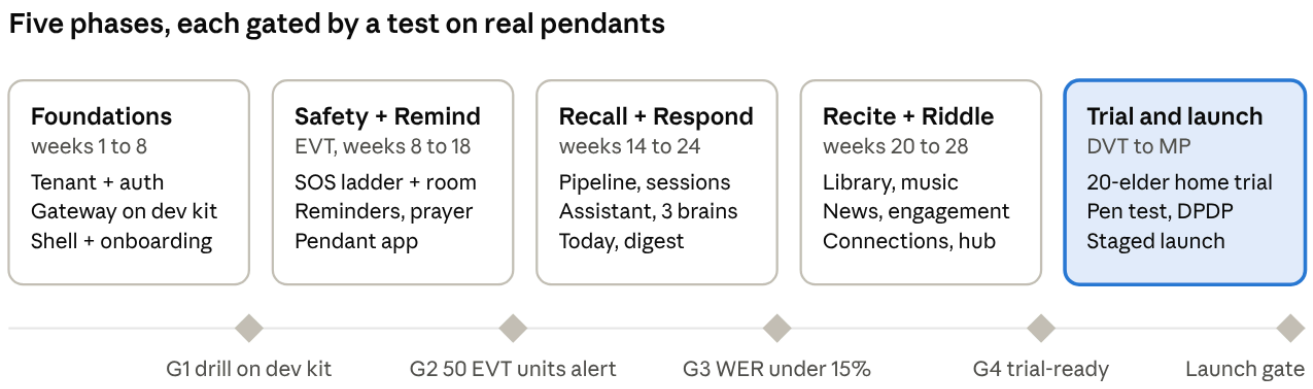
Event	Properties	Purpose
note_sent, note_played, checkin_answered	role	Engagement
battery_low, not_charged_3d, offline_2h		Device health
provider_fallback, cap_reached	provider, engine	Provider reliability and plan sizing

Success metrics (90 days after launch)

Metric	Target
Households completing onboarding within 48 h of unboxing	80%
Wearer voice turns per day (median household)	5 or more
Days per week a family member opens Today	4 or more
Important-reminder acknowledgement rate	85% or more
Median time to acknowledgement in drills and real incidents	Under 2 min
False fall alerts per wearer per month	1 or fewer
Word error rate on the golden set per supported language	15% or lower
Summary edits per session	0.3 or fewer
Households with recording consent given	70% or more
30-day household retention	90% or more
Support tickets per household per month	0.2 or fewer
Provider cost per household per month	Within the plan allowance for 90% of households

20. Release plan, dependencies & open questions

Software runs ahead of hardware: the Gateway and tenant are built against a Nordic development kit in the first eight weeks (spec §9), safety and reminders are proven on the 50 EVT units, and the full V1 is what the 20-elder DVT home trial uses before mass production.



software release plan · five phases against the hardware programme

Each gate is a test on real pendants, not a date: a phase does not close until its gate passes.

Gate criteria

Gate	Passes when
G1	A drill on the dev kit reaches push, SMS and call to a test ladder in under 2 minutes; tenant provisioning under 90 s in India and EU
G2	50 EVT units run the SOS and reminder flows for two weeks with zero lost events; the out-of-range late-alert path proven
G3	Word error rate under 15% on the golden set for Hindi, Tamil and Indian English with each of the three Brains; summary edit rate under 0.3
G4	Every section of this document demonstrable end to end; accessibility audit passed; onboarding under 25 minutes for five test families
Launch	Home-trial retention over 90%; pen test closed; DPDP notice and consent flows reviewed; certifications complete and yield over 95% on the hardware side

Dependencies on firmware and the Gateway

- Firmware: the compact-schedule slot count, event ids on every notification, resumable file pull by offset, the "memory nearly full" spoken line, cancelled-fall reporting, a mute-state event, and the drill flag on SOS events.
- Gateway: background BLE with state restoration on iOS and a foreground service on Android, multi-bond with hub priority, one-SMS fallback on Android, WebRTC for the voice line, and a webview shell that passes safe-area and keyboard events to the web app.
- Commercial: telephony provider with India DLT templates, music-service developer agreements, a news-source list per language, a panchang and prayer-calculation source, and the provider contracts with zero-retention terms.

After V1

Version	Adds
V1.1	Apple Watch and Wear OS companion, WhatsApp alert channel, licensed wake word, care-professional share links (time-boxed read access for a doctor), repetition counts in the digest
V2	Bring-your-own server (export plus self-host of the tenant image), Apple Find My and Google Find Hub, third-party skills and an app store on the desktop, Lite+ hardware features when the enclosure gains Wi-Fi

Open questions

- ☐ Panchang and festival data source per region, and which prayer-calculation methods ship in India by default
- ☐ Compact-schedule slot count and whether recurrence rules can live on the pendant instead of expanded occurrences
- ☐ Plan allowances: transcription minutes and assistant turns per month, and the price of continuous mode
- ☐ Whether to offer always-on location sharing to the Owner in Wearer-managed households at V1 or hold it to V1.1
- ☐ Voice-line (two-way audio through the pendant) at V1, or V1.1 if the Gateway WebRTC work slips
- ☐ YouTube Music scope given no official playback API: search-and-open only, or drop it from V1 marketing
- ☐ Default reading language of summaries when family members read in different languages

- ☐ Whether Viewers may receive incident pushes by default or only by opt-in
- ☐ Legal review of the consent records as evidence under DPDP and two-party-consent jurisdictions
- ☐ Support model: in-app chat with Svaraa first, then human, and the hours for the India launch